



World Scientific News

An International Scientific Journal

WSN 190(1) (2024) 1-69

EISSN 2392-2192

Navigating the Cyber security Landscape: A Comprehensive Review of Cyber-Attacks, Emerging Trends, and Recent Developments

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ABSTRACT

The exponential growth in internet usage has reshaped daily transactions, prompting individuals and companies to increasingly engage in cyberspace rather than traditional real-world settings. This shift has been notably accelerated by factors such as the COVID-19 pandemic. The widespread adoption of the digital environment has led to a shift in criminal activities, with traditional crimes now extending into the digital space. Cybercrime has become a significant concern as criminals exploit vulnerabilities in the online world. The emergence of technologies like cloud computing, Internet of Things (IoT), social media, wireless communication, and cryptocurrencies has heightened security concerns in cyberspace. The trend of cyber criminals offering cyber attacks as a service reflects a concerning shift toward automation for broader impact. Exploiting vulnerabilities across hardware, software, and communication layers amplifies the potential impact of these attacks, emphasizing the need for robust cybersecurity defenses. The landscape of cyber threats encompasses various types of attacks. These include distributed denial of service (DDoS), phishing, man-in-the-middle, password attacks, remote attacks, privilege escalation, and the use of malware. The evolving landscape of cyber threats and advanced evasion techniques has rendered traditional protection systems, including firewalls, intrusion detection systems, antivirus software, and access control lists, less effective in detecting sophisticated attacks. Addressing the urgent need for innovative and effective solutions to prevent cyber attacks is crucial. Reviewing recent attacks, understanding attack patterns, and exploring detection techniques are

essential steps in staying ahead of cyber threats. The article's discussion of both technical and non-technical solutions for early recognition is key to developing a comprehensive and proactive cybersecurity framework. Leveraging trending technologies like machine learning, deep learning, cloud platforms, big data, and block-chain holds promise as a solution for addressing current and future cyber attacks. The mentioned technological solutions, including machine learning and deep learning, can play a crucial role in various aspects of cybersecurity. They offer capabilities for detecting malware, intrusion detection, spam identification, DNS attack classification, fraud detection, recognizing hidden channels, and distinguishing advanced persistent threats, enhancing the overall defense against sophisticated cyber attacks. While machine learning and deep learning show promise in cybersecurity, their susceptibility to evasion techniques is a critical consideration. Developing robust solutions requires addressing the challenges posed by intelligent cyber attacks and continuously evolving evasion methods.

Keywords: Cyber security, cyber attacks, cyber threats, emerging trends, network security, threat intelligence, security solutions

1. INTRODUCTION

The Internet, initially a platform for communication and sharing, rapidly transformed the global landscape, influencing the geography and connectivity of the entire world. The 21st century stands out as the era where global geography is intricately intertwined with the expansive network of the Internet. The high-speed communication facilitated by the Internet has forged robust connections among people globally. This connectivity has created strong bonds between states, impacting critical domains such as commerce, politics, economics, and socio-cultural interactions (Pan et al. 2011). The Internet is comprised of three fundamental entities: computers, users, and networks. The evolution of network technologies is closely tied to the dynamic advancements in computer technologies and the diverse capabilities of user segments (Pan et al. 2011). The synergy among these elements has been instrumental in shaping the ever-changing landscape of the Internet and its associated technologies. The widespread adoption and development of network technologies have introduced critical security challenges. Recognizing the growing cybersecurity challenges, there have been concerted efforts to establish a secure cyber environment safeguarding the assets of institutions, organizations, and individuals (Safa et al. 2016).

While the term “cyber” is often used to describe networks and information systems (Von Solms and Van Niekerk. 2013), it's worth noting that the concept of “virtual reality” typically refers to an immersive simulated environment created using computer technology. Cybersecurity plays a crucial role in safeguarding the security, integrity, and confidentiality of communication, life, integration, as well as tangible and intangible assets and data within the electronic environments established by institutions, organizations, and individuals using information systems. Cybersecurity serves to ensure the security of virtual life within cyber networks, encompassing measures to protect against threats and vulnerabilities in the digital domain. Cyber security is responsible for safeguarding the infrastructure of information systems, ensuring the protection of data integrity, and preserving confidentiality (Craig et al. 2014). The primary objective of cybersecurity is to ensure the security and protection of data belonging to both individuals and institutions on the Internet. Overlooking the importance of cybersecurity can lead to serious threats.

The consequences of neglecting cybersecurity are significant. Malicious actors can exploit vulnerabilities, infiltrate devices through networks, and potentially hijack data or steal sensitive information like credit card details and user ID passwords (Wang and Lu. 2013). Cyberattacks have the potential to inflict substantial financial damage on individuals, institutions, large corporations, and even state governments. Recent studies highlight the significant economic impact, with cyberattacks costing billions of dollars to the global economy. Contemporary cyber attacks are not confined to simple computer breaches; they extend to large-scale operations targeting major businesses and even state governments. Robust cybersecurity policies are essential to prevent and mitigate the impact of these sophisticated attacks, emphasizing the importance of proactive measures to safeguard digital assets and infrastructure (Papp et al. 2015).

Considering all these mentioned aspects together, in this study, we aim to conduct a comprehensive analysis explaining the basics and importance of cybersecurity is highly relevant and crucial. In this comprehensive study, covering all aspects of cybersecurity, discussing shared risks and threats, and examining solutions for prevention provides valuable insights. A guided study from the fundamentals of cyber and network security to common attacks will undoubtedly contribute to researchers' understanding and preparedness in this critical field.

Creating a list of frequently used phrases and their acronyms is a thoughtful approach to enhance the transparency and precision of your paper's language (**Table 1**). The review paper, aiming to be the most comprehensive article on cybersecurity, brings a distinctive approach by covering various perspectives. Unlike previous survey papers that often focused on specific subjects, such as threats, attacks, block-chain technology, machine learning techniques, challenges, or historical development, your paper seeks to provide a more holistic view of the cybersecurity landscape. The distinctive aspect of our study lies in its detailed exploration of various facets of cybersecurity. Our method of breaking down the cybersecurity problem into smaller pieces and extensively covering each aspect in different sections is commendable (Maglaras et al. 2018; Dasgupta et al. 2020; Taylor et al. 2020; Yaacoub et al. 2021; Kaur and Ramkumar. 2022; Khan et al. 2023; Waseem et al. 2023).

Table 1. Some commonly used Context and their Acronym

Sl. No.	Acronym	Context
1	AAA	Authentication, Authorization, and Accounting
2	ACL	Access Control List
3	ARP	Address Resolution Protocol
4	ARPANET	Advanced Research Projects Agency Network
5	AWS	Amazon Web Services
6	AFP	Apple Filling Protocol

7	BPDU	Bridge Protocol Data Unit
8	CCS	Change Cipher Specification
9	CTSS	Compatible Time-Sharing System
10	CNAs	Computer Network Attacks
11	CIA	Confidentiality, Integrity, and Availability
12	CAM	Content Access Memory
13	COVID-19	Corona virus
14	XSS	Cross-Site Scripting
15	DoD	Department of Defense
16	DDoS	Distributed Denial of Service
17	DNP3	Distributed Network Protocol 3
18	DNS	Domain name System
19	DHCP	Dynamic Host Configuration Protocol
20	ESA	Email Security Appliance
21	GPL	General Public License
22	GARP	Gratuitous ARP
23	HTML	Hyper-Text Markup Language
24	HTTP	Hyper-Text Transfer Protocol
25	HTTPS	Hyper-Text Transfer Protocol Secure
26	ICA	Independent Computing Architecture
27	ICMP	Internet Control Message Protocol
28	IMAP	Internet Message Access Protocol
29	IoT	Internet of Things
28	IP	Internet Protocol
29	IPsec	Internet Protocol Security

30	IPv4	Internet Protocol Versions 4
31	IPv6	Internet Protocol Versions 6
32	IDPS	Intrusion Detection, Prevention, and Protection systems
33	LPP	Lightweight Presentation Protocol
34	LDA	Linear Discriminate Analysis
35	LAN	Local Area Network
36	ML	Machine Learning
37	MTA	Mail Transfer Agent
38	Malware	Malicious Software
39		
40		
41		
42		
43	MBSA	Microsoft Baseline Security Analyzer
44	MAC	Media Access Control
45	NDR	Network Data Representation
46	NTP	Network Time Protocol
47	NCP	Net-Ware Core Protocol
48	ISO	International Organization for Standardization
49	OSI	Open Systems Interconnection
50	PAP	Password Authentication Protocol
51	P2P	Peer-to-Peer
52	PIN	Personal Identification Number
53	POP3	Post Office Protocol 3
54	PCA	Principal Component Analysis

55	QoS	Quality of Service
56	RTCP	Real-Time Transport Control Protocol
57	RPCs	Remote Procedure Cells
58	SSL	Secure Socket Layer
59	SDL	Security Development Lifecycle
60	SLIP	Serial Line Internet Protocol
61	SSID	Service Set Identifier
62	SCP	Session Control Protocol
63	SMTP	Simple Mail Transfer Protocol
64	SNMP	Simple Network Management Protocol
65	SDN	Software Defined Network
66	STP	Spanning Tree Protocol
67	SQL	Structured Query Language
68	SCADA	Supervisory Control and Data Acquisition
69	SVM	Support Vector Machine
70	SYN	Synchronize
71	TCP RST	TCP Reset
72	TELNET	TELEtype NET-work
73	TLD	Top-Level Domain
74	FTP	File Transfer Protocol
75	TCP	Transmission Control Protocol
76	TLS	Transport Layer Security
77	TCSEC	Trusted Computer System Evaluation Criteria
78	UDP	User Datagram Protocol
79	VLAN	Virtual LAN
80	VMM	Virtual Machine Monitor

81	VPN	Virtual Private Network
82	WSA	Web Security Appliance
83	WEP	Wired Equivalent Privacy
84	WSNs	Wireless Sensor Networks

This approach allows for a thorough understanding by detailing attack vectors, proposing possible remedies for each attack class, and addressing the associated challenges. The approach of revisiting similar information across sections while emphasizing different aspects enhances the depth and richness of your study. The nuanced perspectives and more detailed information in certain sections contribute to a well-rounded and thorough exploration of the various dimensions of cybersecurity, providing readers with a comprehensive understanding of the subject matter. This paper makes a valuable contribution to both researchers and individuals seeking to enhance their understanding of cybersecurity, spanning from fundamental to advanced levels. The summarized main topics covered in the article provide a clear roadmap for readers to explore and grasp the diverse aspects of cyber-security comprehensively.

2. CYBER SECURITY FUNDAMENTALS

In the “Cyber Security Fundamentals” section, this paper delves into the history of cyber crime, outlines security fundamentals, explores basic principles of potential cybersecurity solutions, analyzes the reasons behind the increase in cyber attacks, underscores the importance of cybersecurity, examines attacks from technical and non-technical viewpoints, and suggests measures to address these challenges (Von Solms and Van Niekerk. 2013). Cybersecurity is crucial in safeguarding digital systems. The rise in cyber attacks can be attributed to evolving technologies, increased connectivity, and the growing value of digital assets.

History of cyber crime and cyber security

Various terms like data security, information security, network security, and cybersecurity are often used interchangeably (Denning. 1982). Data security specifically focuses on safeguarding digital data at every stage of its existence, preventing unauthorized access, modification, or disclosure. It forms a crucial aspect of the broader field of cybersecurity, which encompasses the protection of digital systems, networks, and information against a wide range of cyber threats. Information security involves a comprehensive set of practices aimed at preventing unauthorized access, use, disclosure, modification, review, recording, or destruction of both physical and electronic information (Blackley et al. 2004). The fundamental goals of information security revolve around the CIA triad: Confidentiality, Integrity, and Availability. It aims to ensure that data is kept confidential and only accessible to authorized individuals, that data remains unaltered and maintains its accuracy and reliability (integrity), and that information is available when needed without unauthorized disruption (availability). Network security is dedicated to preserving the confidentiality, integrity, and accessibility of computer networks and the data transmitted through communication channels

(Denning. 1982). Cybersecurity is a broader concept that encompasses the protection of not only computers, servers, and electronic systems but also mobile devices and entire computer networks. It involves the practice of safeguarding digital assets, including data, from various forms of malicious attacks, such as cyber threats, hacking attempts, and other cybercrimes. Data security, information security, and network security address specific aspects, while cybersecurity provides a holistic approach, safeguarding end-to-end information flows across diverse digital landscapes (Cole. 2011). In contemporary discussions, “cybersecurity” has indeed become the prevailing term, reflecting the comprehensive nature of efforts to protect against a wide range of cyber threats.

Cybercrime has a history that extends over several years. In the early stages, defending the digital world was relatively easier due to a smaller number of machines and less sophisticated attacks. As technology has advanced and our digital interconnectedness has grown, cyber attacks have become more intricate and widespread, posing greater challenges for cybersecurity measures. Technological advancements have provided cybercriminals with tools to automate and execute more sophisticated attacks. The expanding Internet landscape, encompassing smart phones, tablets, IoT devices, cloud platforms, and social media, among others, has increased the attack surface. The evolution of cybercrime from simple pranks to sophisticated attacks has had significant economic consequences, costing trillions of dollars globally each year (Aslan and Yilmaz. 2021). The landscape of cyber crimes has diversified over the decades, leading to classifications that encompass a wide range of malicious activities (Table 2). These classifications include but are not limited to hacking, identity theft, financial fraud, ransomware, and cyber espionage.

In the early 1940s, when the first computers emerged, the absence of internet connectivity and limited computer usage meant there were no threats or attacks on computers (Cyber security. 2023). Phone phreaking emerged in the 1950s, where enthusiasts manipulated phone system protocols to make free calls or reduce long-distance fees. During this era, certain phone companies struggled to prevent these activities. The techniques employed by phone phreaks foreshadowed aspects of future hacking methods that would be adapted to exploit vulnerabilities in computer systems as technology advanced.

The term “hacking” in the context of computer systems is believed to have originated in the 1960s. In 1965, the discovery of the first vulnerability in the IBM 7094 Compatible Time-Sharing System (CTSS) machine marked an early instance in the history of computer security (Walden. 2011). In 1967, IBM hired a group of students to explore their newly designed computer, the IBM System/360 Model 91. This initiative was part of IBM’s Student Loan Equipment Program, allowing students to gain hands-on experience with the cutting-edge technology. The exploring the IBM System/360 Model 91 in 1967 and gaining access to various system parts exemplifies an early form of ethical hacking, revealing vulnerabilities in computer systems for constructive purposes (Cutter et al. 2012). The foundation of cyber-security indeed took a significant step forward in the early 1970s with ARPANET, the first packet-switched network that predates the modern Internet. In 1971, Bob Thomas created the first computer virus known as the “Creeper.” The Creeper was an experimental self-replicating program that could move between computers on the ARPANET network, displaying a message saying “I’m the creeper, catch me if you can!” This marked an early example of a computer program designed to spread autonomously between computers, laying the groundwork for the concept of computer viruses. Following the creation of the “Creeper” virus, Ray Tomlinson developed the “Reaper” in 1971.

The Reaper was essentially an early form of antivirus program designed to move across the ARPANET, identify systems infected with the Creeper virus, and then delete the Creeper. “Reaper” was an early example of an antivirus program, as it was designed to detect and remove the “Creeper” virus (Choi et al. 2023). As for Kevin Mitnick, he gained notoriety as a hacker, and in 1979, he was arrested for the first time for engaging in cybercriminal activities. Mitnick’s subsequent activities and legal troubles played a role in shaping public awareness of computer security issues (Choi et al. 2023).

The 1980s witnessed a surge in computer-related attacks, with a notable emphasis on the use of computer viruses during this decade. The term “cyber espionage” started gaining prominence, reflecting concerns about threats from other governments. In 1985, the U.S. Department of Defense (DoD) established computer security guidelines known as the “Trusted Computer System Evaluation Criteria” (TCSEC), later named “Orange Book” (Russell et al. 1991; Lehtinen et al. 2006). In 1986, German hacker Marcus Hess infiltrated government systems in the USA, East Asia, and Europe, gaining access to approximately 400 military computers (Li and Liu. 2021). The compromised information encompassed sensitive technologies related to space, satellites, and aircraft. This incident, among others, highlighted the growing need for heightened security measures (Reddy and Reddy. 2014). In response to the escalating threat landscape, commercial antivirus software was first released in 1987, underscoring a proactive step towards protecting computer systems from emerging cyber threats (Conti et al. 2018). The 1990s witnessed significant growth in computer systems and the internet, leading to increased connectivity. During this time, computer viruses and their various iterations gained popularity. In 1996, macro-viruses, a type of virus that infects documents and uses macro programming scripts, emerged. Towards the late 1990s, notable viruses like Melissa and ILOVEYOU made headlines by infecting millions of computers worldwide. In 1995, Netscape introduced the Secure Sockets Layer (SSL) protocol, a pioneering development in internet security (Aslan et al. 2021). SSL provided a secure way to establish encrypted connections over a computer network, enhancing the privacy and integrity of data transmitted between users and websites.

During the 2000s, the internet experienced exponential growth, and computers became ubiquitous in both work and home settings. This widespread adoption of computers, while enhancing productivity, also brought about increased security risks. The escalating usage of computers coincided with a rise in cybercrime. Notably, in the 2000s, the emergence of the first organized hacker groups and the popularity of computer worms and Trojans as attack methods marked a shift in the landscape of cyber threats. In the mid-2000s, the landscape of cyber threats evolved, and users faced risks by simply visiting infected websites, known as drive-by downloads. In 2004, the My-doom worm demonstrated the capability for distributed denial of service (DDoS) attacks and remote access, showcasing the potential for multifaceted cyber threats (Aslan et al. 2021). Subsequently, in 2007, the Zeus Trojan introduced sophisticated techniques like drive-by downloads and spam emails to steal login credentials.

In the 2010s, cybercriminals exploited security vulnerabilities in software and computer network protocols, leading to significant breaches. These incidents resulted in financial losses for individuals in the millions and had major companies and even countries facing annual losses in the billions. One notable event in 2016 was the Mirai malware, which capitalized on vulnerabilities in Internet of Things (IoT) devices. Mirai orchestrated large-scale Distributed Denial of Service (DDoS) attacks by leveraging a botnet of compromised IoT devices (Al-Hadhrami and Hussain. 2021).

Ransomware-related attacks witnessed a surge in popularity between 2010 and 2020. The WannaCry ransomware, in 2017, notably encrypted computer systems globally, affecting 150 countries. Similarly, the LockerGoga ransomware caused significant damage by blocking infected systems and resulting in millions of dollars in losses (Kumar and Carley, 2016; Kumari and Jain, 2023). In 2020, the CovidLock ransomware added a new dimension by targeting Android devices, encrypting data and denying access (Rai et al. 2022). The 2020s have indeed seen an increase in the sophistication and availability of hacking tools and services (Pauli. 2013). Professional websites offering hacking tools as a service have become more prevalent, making it easier for Individuals with malicious intent to carry out cyber attacks. The motivation behind cyber attacks has evolved to include not just mischief but also financial gain (Manky. 2013). Timely and effective cyber attacks can yield substantial profits, leading to significant investments by both big companies and governments in cyber security measures (Aslan and Samet. 2020). This ongoing evolution emphasizes the importance of continually adapting and enhancing defenses to counter the dynamic landscape of cyber threats (**Table 2**).

Table 2. Cyber Crimes / Year

Cyber Crimes	Year
Achieving a year without computer crime would require collaborative efforts in cyber-security, education, and law enforcement to address vulnerabilities.	1940
Phone phreaking , a practice exploring and manipulating telephone systems	1950
Hacking involves exploiting vulnerabilities in computer systems, while vulnerability refers to weaknesses that could be exploited.	1960
Computer security as a field emerged with the rapid growth of computer systems.	1970
ARPANET, the precursor to the internet.	1980
Computer viruses and worms gained popularity.	1990
The Internet has experienced exponential growth since its inception.	2000
Cybercriminals continually exploit security breaches in computer systems.	2010
Cybercrime has indeed evolved into a sophisticated and lucrative industry.	2020

The methods employed to spread cyber attacks have evolved over time, reflecting the dynamic nature of cyber threats. Vulnerabilities in hardware, software, and networks remain common targets for exploitation. Additionally, phishing scams, where attackers deceive individuals into revealing sensitive information, and social engineering techniques, which

manipulate human psychology to gain unauthorized access, have become prevalent strategies in the cyber threat landscape. Cyber attacks often leverage various channels for propagation. Drive-by downloads, where malware is automatically downloaded onto a user's device without their knowledge or consent, pose a significant threat. Malicious email attachments remain a common vector, with attackers sending deceptive emails containing harmful payloads. Fake applications, especially on mobile platforms, serve as another avenue for spreading malware (**Table 3**). The evolution of attack tools has empowered cybercriminals to carry out a variety of malicious activities. This encompasses gaining unauthorized access to individuals' banking systems, stealing sensitive data from major corporations, encrypting data on computer hard drives, and disrupting access to significant company resources through Distributed Denial of Service (DDoS) attacks. The financial impact of cyber attacks has reached staggering figures, costing billions of dollars annually worldwide. The proliferation of new devices, including smartphones and Internet of Things (IoT) devices, has expanded the attack surface. With each new connected device, there is an increased potential for vulnerabilities, creating additional challenges for cybersecurity efforts. Cybercriminals continually refine and evolve existing cyber attacks by generating different versions and developing new attack variants tailored for smart phones and IoT devices. Recent studies indeed highlight a rising trend in fake applications, backdoors, and banking Trojans specifically targeting mobile devices (Aslan and Samet. 2020). Cyber attacks have expanded to target various emerging technologies and digital environments. Social media platforms, Internet of Things (IoT) devices, crypto currency systems, and cloud computing environments are all experiencing an increase in cyber threats.

Table 3. The evolution of cyber attacks over the years

Cyber-Related Attack	Attack Spread Method	Consequences	Year
Vladimir Levin's Attack the Citibank1	Unknown	Instances where around \$ 10 million is stolen could involve various cybercrimes like large-scale hacking, financial fraud, or ransom-ware attacks.	1994-1995
Melissa Virus	Exploiting users' trust through phishing emails with malicious attachments is a common tactic employed by cybercriminals.	Numerous countries have experienced substantial financial losses due to various cybercrimes, including large-scale data breaches, ransom-ware attacks, and financial fraud.	1999
ILOVEYOU Worm3	Exploiting users' trust through	Instances where more than 45 million	2000

	phishing emails with malicious attachments is a common tactic employed by cybercriminals.	computers are infected often indicate large-scale malware outbreaks or widespread cyber attacks.	
My Doom worm ⁴	Using attention-grabbing subjects in emails, such as errors or tests, is a common tactic in phishing attacks.	Launching Distributed Denial of Service (DDoS) attacks by allowing remote access is a concerning cyber-security threat. This tactic involves compromising devices to create a botnet, which is then used to flood a target with traffic, disrupting its online services.	2004
Zeus Trojan	Spam emails with drive-by downloads pose a serious threat by delivering malicious content when users click on seemingly harmless links or download attachments.	The theft of login details, including email and bank account credentials, is a serious cyber-security issue often associated with phishing, malware, or data breaches.	2007
Stuxnet Worm	Attacking the programmable logic unit (PLU) by stealing source codes is a significant security concern, especially in industries relying on programmable logic controllers (PLCs).	Gaining control of industrial processes, often through cyber attacks, poses severe risks to critical infrastructure. Such incidents could result from exploiting vulnerabilities in industrial control systems (ICS).	2010
Attack on USA Natural Gas Pipeline	Accessing confidential information through phishing is a	The theft of security credentials poses a significant security risk, potentially leading to	2012

	prevalent cyber threat.	unauthorized access and compromise of sensitive information.	
Power System Real Time Digital Medium Simulator (RTDS)	Real-Time Digital Simulator measurement attacks or control attacks can be detrimental, impacting critical infrastructure or processes.	The Power Cyber CPS (cyber-physical system) security test bed is likely a simulated environment designed to assess and enhance the security of power systems with integrated digital components.	2015
Mirai Malware	The exploitation of vulnerabilities in IoT devices underscores the importance of ensuring robust security measures for these interconnected devices.	The launch of DDoS attacks can disrupt online services by overwhelming them with a flood of traffic	2016
WannaCry Ransomware	The exploitation of windows vulnerabilities is a serious cyber-security concern.	It is a global ransomware attack. Such incidents involve malicious actors encrypting computer hard drives and demanding a ransom for the decryption key.	2017
Emotet Trojan	The influx of emails in the form of spam and phishing campaigns poses a significant cyber-security threat.	The theft of sensitive information, such as credit card details, is a serious security breach that can lead to financial losses and identity theft.	2018
MyFitnessPal	Exploiting software vulnerabilities poses a significant cyber security risk.	The impact on 150 million users suggests a large-scale cyber-security incident, possibly a	2018

		data breach or a widespread attack.	
OMNET Smart Grid (PSCADA)	DDos attack /FDI attack can disrupt the normal functioning of a system or network by overwhelming it with a flood of traffic.	The lack of thorough testing and scenarios is a common vulnerability in cyber security. Comprehensive testing, including penetration testing and scenario simulations, is crucial to identify and address potential weaknesses in systems and processes.	2019
Ransomware attack on Magellan	Receiving emails in the form of spam and phishing campaigns is a common cyber-security threat.	The theft of health data from 365,000 patients is a severe breach with potential privacy and security implications.	2020
CovidLock Ransomware	Exploiting users' trust through the use of COVID-19 statistics in phishing campaigns is unfortunately a tactic that emerged during the pandemic.	The encryption of data on Android devices, coupled with denial of access, could indicate a ransomware attack or a security measure implemented by the device owner.	2020
Accellion Supply Chain Attack	Exploiting third-party vulnerabilities is a common tactic in cyber attacks.	The theft of confidential data from large organizations is a serious security breach with potential far-reaching consequences.	2021
Kaseya Ransomware Attack	The use of zero-day exploits indicates a highly sophisticated and concerning cyber security threat.	The compromise of data from around 1500 companies with ransom demands ranging from \$50,000 to \$5 million	2021

		per victim is indicative of a large-scale and coordinated ransomware attack.	
Hackivist Attack	Hackivist attack often includes website defacement, distributed denial-of-service (DDoS) attacks, or unauthorized access to system.	Engaging 7.25 million bots in a bot attack suggests a large-scale and potentially orchestrated cyber operation.	2022
MOVEit Attack	The most up-to-date information on manage file transfer specific attack method.	Instances where around \$ 17.5 million is stolen could involve various cybercrimes like hacking, or ransomware attacks.	2023

Principle of information security

The three dimensions of cybersecurity encompass various aspects of protection (**Figure 1**). The first dimension revolves around safeguarding information and is often referred to as the "principle of information security." This principle is grounded in the fundamental concepts of confidentiality, integrity, and availability, commonly abbreviated as CIA (Whitman and Mattord. 2011).

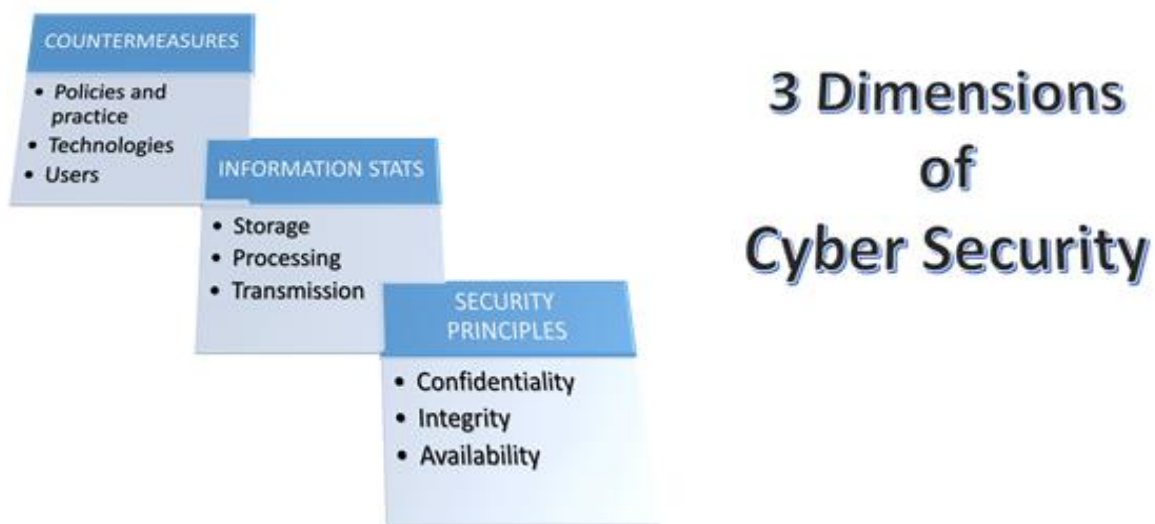


Figure 1. Three Dimensions of Cyber Security

The second dimension of cybersecurity focuses on protecting data in all states—whether it's in storage, transit, or in process (Almadahkah. 2016). Data can be transferred through various means, including Sneaker net, wired networks, and wireless networks. In this context, cybersecurity measures must ensure the confidentiality, integrity, and availability of data. Whether data is at rest, in motion, or being processed, robust security measures are essential to safeguard against potential threats. The third dimension of cybersecurity extends beyond the technical aspects and involves the implementation of additional tools. This includes the formulation and enforcement of policies and practices, the integration of new technologies, and raising user awareness. By incorporating these elements, cybersecurity efforts become more holistic, addressing not only technological vulnerabilities but also human factors and governance. This comprehensive approach is crucial for creating a resilient cybersecurity framework that can adapt to the evolving landscape of cyber threats in cyberspace.

Confidentiality

Confidentiality in the digital world involves safeguarding information from unauthorized access by users and programs. The classification of data into categories like Top Secret, Secret, Confidential, and Unclassified helps define the level of sensitivity and the corresponding security measures required (Schumacher et al. 1999). Top Secret and Secret data are deemed extremely crucial, and their protection is of paramount importance due to the potential for significant damage to national security if disclosed. Consequently, access to such data must be rigorously restricted to authorized individuals or entities. Confidential data, while not as sensitive as Top Secret or Secret, still holds importance, and its disclosure should be avoided. Unclassified data is not sensitive and is considered publicly available, posing no impact on national security. However, it's crucial for governments, organizations, and companies to prioritize training their employees to safeguard all types of information, including Top Secret, Secret, and Confidential data. To maintain the confidentiality of data, encryption, authentication, and access control techniques are used.

Encryption is a mathematical technique crucial in cybersecurity. It transforms original information into an unreadable version, concealing it from unauthorized access (Puttaswamy et al. 2011). This process involves using algorithms and cryptographic keys to encode data, making it challenging for unauthorized individuals to decipher without the corresponding decryption key. Encryption has proven to be a fundamental tool in cybersecurity, providing protection against various threats such as brute-force attacks, spyware, and ransomware. Access control is a fundamental security concept that plays a crucial role in minimizing the risk of unauthorized access to computer systems and networks. It encompasses various protection mechanisms designed to ensure that only authorized individuals or systems have the appropriate permissions to access specific resources or data. Access control mechanisms include user authentication, authorization processes, and policies that dictate the level of access granted to different users or entities. The concept of AAA, which stands for Authentication, Authorization, and Accounting, forms a fundamental framework in cyber-security. Authentication is the process of verifying the identity of users or systems to ensure that they are who they claim to be. This often involves the use of a username and password combination, but it can extend to more advanced methods such as biometrics, multi-factor authentication, or cryptographic keys (Metz. 1999). By authenticating users, organizations prevent unauthorized access and establish a secure foundation for subsequent security services. Authorization determines the level of access granted to authenticated users. It specifies which computer and network resources users

can access and what operations they are allowed to perform (Paolini et al. 2020). This step ensures that users have appropriate permissions based on their roles or specific attributes, minimizing the risk of unauthorized actions. Accounting involves the monitoring and recording of users' activities, detailing the resources they access and any changes they make. This provides an audit trail, allowing organizations to review and analyze user actions for security and compliance purposes. Accounting contributes to a comprehensive understanding of system activities and helps in detecting any unusual or potentially malicious behavior.

Integrity

Integrity in the context of cybersecurity is closely tied to the accuracy, quality, consistency, and completeness of data throughout its entire life cycle (Sivathanu et al. 2005). Ensuring data integrity means that information can only be modified by users with the appropriate permissions to do so. This principle is crucial for maintaining the reliability and trustworthiness of data. Modern enterprises prioritize data integrity for various reasons, including: Reliable and accurate data is essential for making informed decisions and conducting business operations (Pipino et al. 2004). Many industries and regulatory frameworks require organizations to maintain data integrity to meet legal and compliance standards. Customers and stakeholders trust organizations that can demonstrate the integrity of their data, fostering confidence in the information provided. Data integrity is vital for several aspects of data management. The ability to recover data accurately and completely, especially after an incident or failure. The ability to track changes made to data, providing a historical record of modifications. Maintaining the reliability and consistency of data across interconnected systems and devices. Various factors can compromise data integrity, including physical device compromise, disk crashes, malware, hacking, inconsistent data formats, errors during data transfers, and human mistakes. Using cryptographic hash functions to generate a fixed-size string of characters (hash) based on the content of data. Any change to the data results in a different hash, enabling quick detection of alterations. Verifying that data remains consistent and accurate across different parts of a system or database. Ensuring that data adheres to predefined rules and standards, identifying any inconsistencies or errors. By implementing these measures, organizations can mitigate the risks to data integrity and maintain the reliability of their information assets.

Availability

Availability in the context of cybersecurity is closely tied to data accessibility. It signifies that data should be accessible to authorized users when they request or need to use them (Whitman and Mattord. 2013). System failures and cyber attacks pose significant threats to availability, potentially leading to disruptions or denial of service. Technical issues, hardware malfunctions, or software glitches that lead to system downtime. Deliberate efforts, such as Distributed Denial of Service (DDoS) attacks or other forms of cyber threats, intended to disrupt or deny access to information systems and services. Ensuring availability involves implementing strategies to mitigate the impact of such factors, including redundancy in systems, disaster recovery plans, and security measures to defend against cyber attacks. Distributed Denial of Service (DDoS) attacks represent a severe threat to the availability of a system by overwhelming it with a flood of traffic, making it difficult or impossible for legitimate users to access the services. While there is no foolproof method to completely stop

DDoS attacks, several strategies can be employed to mitigate their impact and ensure availability regularly backing up critical data and system configurations to facilitate recovery in the event of an attack. Implementing redundant systems and resources to maintain functionality even if one part of the system is compromised. Designing systems to be resilient in the face of disruptions, ensuring they can adapt and continue functioning under stress. Up-to-Date Operating Systems and Software keeping software and operating systems current with the latest security updates to patch known vulnerabilities. Designing the system architecture to eliminate or minimize vulnerabilities that could disrupt overall functionality. Early Implementing monitoring tools and systems to detect and respond to failures as soon as they occur, reducing downtime.

Why cyber attacks are increasing

The last few years have witnessed a rapid increase in the importance and utilization of the Internet on a global scale. This surge has resulted in the digital transformation of various aspects of daily life. The COVID-19 pandemic, in particular, played a significant role in accelerating this process, prompting a widespread shift to digital platforms for various activities.

For example, people now develop friendships in social media environments, access online banking systems, and plan online seminars and meetings (Figure 2). The shift of daily activities to the Internet has given rise to cybercrime, orchestrated by individuals or organized groups commonly known as hackers. These hackers possess advanced knowledge of operating systems, quick programming skills, and the ability to identify software and system vulnerabilities swiftly (Aslan. 2022).

Key factors contributing to the proliferation of cybercrime include: Hackers leverage their in-depth understanding of operating systems and programming to exploit weaknesses effectively.

The continuous creation of new attack tools provides cyber attackers with powerful capabilities, making it easier for them to carry out sophisticated attacks. The potential for significant economic gain serves as a strong motivator for cyber attackers. Cybercrime can yield substantial financial benefits, further incentivizing malicious activities. Recent studies highlight the staggering economic impact of cyber attacks, measured in trillions of dollars, and this damage continues to escalate. These factors collectively contribute to the escalating frequency and sophistication of cyber-related attacks. Understanding these motivations is crucial for developing effective strategies to counter and mitigate the evolving landscape of cyber threats. The major reasons for cyber attacks can be categorized as follows (**Figure 2**): Vulnerabilities and weaknesses in existing systems provide opportunities for cyber attackers to exploit and compromise security.

The rapid adoption of new and emerging technologies introduces potential vulnerabilities that cybercriminals may target. As knowledge and expertise in cybersecurity increase, so does the potential for more sophisticated and targeted cyber attacks. The increasing integration of daily activities into the digital realm provides a broader attack surface for cybercriminals. Cyber attacks can originate from anywhere in the world, making it challenging to detect and attribute them to specific locations or entities.

Understanding these reasons is crucial for developing proactive cybersecurity measures to address vulnerabilities, mitigate risks, and protect against the diverse motivations behind cyber attacks.

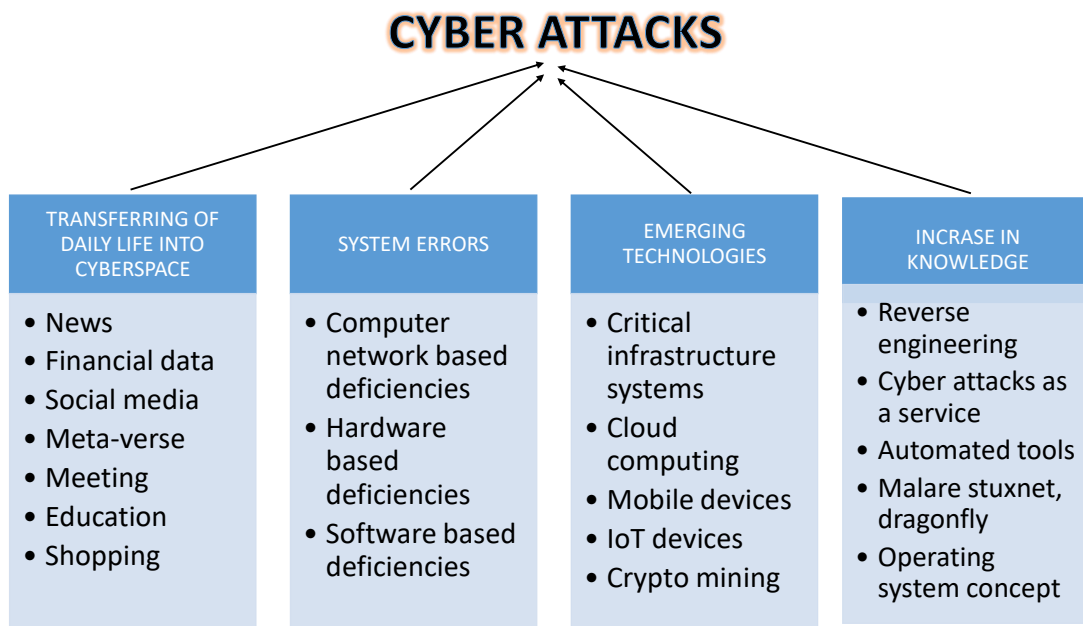


Figure 2. Reasons for cyber attacks

Attacks caused by hardware deficiencies

Hardware-based attacks pose unique challenges as they can exploit vulnerabilities deeply rooted in the physical components of a system. Unlike software, which can be patched or updated, mitigating hardware flaws often requires more extensive measures such as firmware updates or hardware replacements. While Trojan horses are typically associated with software-based attacks, they can indirectly contribute to hardware-related issues (Karri et al. 2010). For instance, certain Trojans might execute processes that lead to increased resource utilization, reduced performance, and even power consumption, potentially contributing to hardware strain or failure. The unauthorized copying of hardware components and acquiring untrusted or counterfeit parts can introduce significant security risks. These actions can lead to the integration of compromised or malicious elements into a computer system, creating potential backdoors for attackers. Ensuring the integrity of hardware sources and utilizing trusted supply chains are essential measures to mitigate the risks associated with illegal copying and obtaining un-trusted components (Karri et al. 2010).

The Intricacies of integrated circuits and the interconnected nature of hardware components indeed make it difficult to detect vulnerabilities. A subtle change in one integrated circuit can have ripple effects across multiple components, and these alterations may not be immediately apparent. Implementing tamper-proof hardware devices, hardware watermarking (Song et al. 2020), and obfuscation techniques are proactive measures to enhance cybersecurity against hardware-related attacks. Tamper-proof devices resist physical manipulation, hardware watermarking helps in verifying the authenticity of components, and obfuscation adds layers of complexity to make it harder for attackers to understand and exploit the hardware (Tehranipoor and Wang. 2011).

Attacks caused by software-based bugs

Software vulnerabilities remain a significant factor in cyber attacks. As technology evolves, so do the complexities of software systems, creating an ever-expanding attack surface. The discovery and exploitation of software errors and vulnerabilities continue to be a common avenue for cyber attackers (McGraw. 2002; Aslan. 2016). Some common causes of software-related vulnerabilities and errors. Failing to properly validate and sanitize user inputs can lead to security vulnerabilities, such as injection attacks. Inadequate management of user permissions and access rights can result in unauthorized access and potential misuse. Flaws in the authentication process may allow unauthorized users to gain access to sensitive information or functionalities.

Issues during data migration processes can introduce errors and vulnerabilities if not handled correctly. Improper handling of data buffers can lead to overflow, enabling attackers to execute arbitrary code and compromise the system. SQL injection occurs when malicious SQL queries are inserted into input fields, exploiting vulnerabilities in database interactions. XSS vulnerabilities allow attackers to inject malicious scripts into web pages, compromising user data and interactions.

Failing to update or replace software components with known vulnerabilities exposes systems to exploitation. Insecure implementation of web services and APIs can lead to unauthorized access and data exposure (Aslan and Samet. 2016). Insufficient testing for security vulnerabilities may leave software susceptible to exploitation.

The software development process, particularly the gap in security testing and developers' awareness of secure coding practices. Rapid development cycles, combined with insufficient security testing, can indeed result in software vulnerabilities. The diversity of platforms and devices further complicates security considerations, as applications must be resilient across various environments. The examples you provided, such as buffer overflows and SQL attacks, underscore the importance of thorough security measures to prevent unauthorized access and data breaches. Improving security in software development involves educating developers on secure coding practices, integrating security testing throughout the development lifecycle, and fostering a security-aware culture within development teams. Regular updates, patches, and monitoring for emerging threats are also crucial to maintaining the security of software applications across different platforms and devices (Ficco et al. 2017).

The importance of a proactive approach to security throughout the entire software development lifecycle. While software updates are essential for addressing vulnerabilities, they may not always provide a complete solution and, in some cases, can introduce new issues. Emphasizing a secure software development process from the outset, including error-free program design and robust requirements, is indeed a fundamental strategy. Training developers in secure coding practices and integrating security measures at every stage of development can significantly minimize the risk of software-based attacks. Microsoft's implementation of the Security Development Lifecycle (SDL) has been a notable success in enhancing the security of its software products.

The mandatory incorporation of SDL steps has demonstrated its effectiveness in reducing errors and vulnerabilities in the software development process (Aslan. 2020). The comparison between Windows Vista, developed with SDL, and Windows XP, which was not subject to SDL, further underscores the impact of secure development practices. The significant reduction in errors and vulnerabilities in Windows Vista highlights the positive outcomes of incorporating security measures systematically throughout the software development lifecycle.

Attacks caused by vulnerabilities in computer networks

Data transfer over the Internet is susceptible to various security risks. Without proper protection measures, hackers may intercept, modify, or even replace transmitted data. Encrypting data during transfer using secure protocols, such as HTTPS, can significantly mitigate these risks by ensuring that the information remains confidential and intact during transit. The reliance on existing computer network protocols and devices without sufficient security considerations can indeed expose systems to various vulnerabilities (Aslan. 2020). The examples you provided, such as vulnerabilities in TCP, IP, ARP, DHCP, and DNS protocols, underscore the potential risks associated with these foundational elements of network communication. Issues like packet exposure and alteration during transport, DNS-related attacks, and the capture of information due to misconfigured network devices pose significant threats. Implementing security measures such as encryption, proper authentication, and thorough configuration reviews can help address these vulnerabilities and enhance the overall security of computer networks.

Cyber security techniques reflect a comprehensive approach to securing computer networks. Encrypting data helps ensure its confidentiality during transmission, mitigating the risk of unauthorized access. Implementing ACLs allows for the control of network access, helping prevent unauthorized users from gaining entry. Virtualization and VLANs provide segmentation, enhancing network security by isolating different parts of the network. Firewalls monitor and control incoming and outgoing network traffic, serving as a barrier against unauthorized access and potential threats (Aslan. 2020). These systems help identify and respond to potential security incidents, enhancing overall network security. WSA solutions help protect against web-based threats, filtering and monitoring web traffic. ESA solutions focus on securing email communications; detecting and preventing email-based threats. VPNs establish secure, encrypted connections over the internet, safeguarding data during transit.

These protocols provide secure communication over a computer network, ensuring data integrity and confidentiality. While these techniques significantly contribute to network security, as you rightly point out, no solution can offer complete protection. Ongoing improvement of existing techniques, development of new ones, and a proactive approach to adapting to emerging threats are essential components of effective network security strategies (Aslan. 2016). The cybersecurity landscape is dynamic, and continuous efforts are required to stay ahead of potential risks.

Causes Arising from Emerging Technologies

The evolving digital landscape, the continual integration of new devices, software, and technologies into the Internet environment, including smart phones, tablets, IoT devices, and cloud technology, indeed expands the attack surface. The rapid generation of application programs and the growing diversity of connected devices contribute to an increased number of potential targets for cyber attacks. Managing the security implications of this dynamic ecosystem requires proactive and adaptable approach. Storing information from various companies on the same physical infrastructure and relying on third-party management introduces potential security challenges. While cloud services offer scalability and convenience, the shared nature of resources raises considerations about data isolation and protection. Addressing these concerns involves robust encryption practices, thorough vetting of cloud service providers' security measures, and implementing stringent access controls.

Increasing number of smart phones

The widespread use of smartphones is indeed remarkable. With approximately 6 billion smartphones in use, it highlights the pervasive integration of these devices into daily life. Nearly one out of every two people using a smartphone underscores the significant impact and influence these devices have on global communication, connectivity, and access to information.

The vulnerability of smartphones to cyber threats is indeed a growing concern. The abundance of personal data stored on these devices, coupled with the proliferation of software applications and the reliance on wireless networks for internet access, makes smartphones attractive targets for attackers. The increase in malicious software designed for smartphones, as highlighted in cybersecurity threat reports from Symantec (Stevens. 2020) and McAfee (Attaran et al. 2019), underscores the evolving landscape of cybersecurity threats. The adaptability of malware originally crafted for traditional computers to smartphones with minimal code changes further emphasizes the need for robust security measures on these devices.

Increasing number of IoT devices

The rapid integration of smart devices into the Internet of Things (IoT) landscape is indeed transforming the digital environment. With various devices like smart glasses, watches, parks, and automation systems collectively forming the IoT, the number of Internet-connected devices is expected to reach around 50 billion in the near future. This surge in IoT devices contributes to a substantial increase in data traffic, posing challenges for network control. The sheer volume of data generated by IoT devices not only complicates network management but also presents a heightened risk for potential cyber attacks.

Increasing usage of cloud computing

Cloud computing has indeed become a transformative technology, offering on-demand services to users. Major cloud service providers like Amazon, IBM, Google, Rackspace, Microsoft, and Salesforce have played a crucial role in popularizing and advancing cloud services. The ease of use, accessibility from various devices and locations, and the flexibility and scalability of services based on user demand contribute to the widespread adoption of cloud technology. The benefits of cloud services, such as offloading maintenance, repair, and updates to providers, contribute to the increasing demand for this technology. However, it's crucial to address the security issues that arise in cloud environments: Companies and institutions may lose control over their data when stored in the cloud, raising concerns about data privacy and security. Using the same physical infrastructure for different companies can introduce potential threats and access issues.

Uncertainty about where data is stored and the lack of visibility into security measures can be significant challenges (Padhy.2011). Vulnerabilities in virtual machine monitors (VMMs), such as Xen, VMware, and Microsoft Hyper-V, pose risks to the integrity and isolation of virtualized environments. Attacks on data during transfer over computer networks remain a persistent threat. Addressing these concerns requires collaboration between cloud service providers and user organizations. Implementing robust technological and managerial measures, enhancing transparency about data storage and security protocols, and staying vigilant against emerging threats are essential steps in mitigating risks associated with storing data in the cloud.

Increase in the number of critical infrastructure systems

These systems, vital for the daily operations of a society, encompass various sectors such as energy production and distribution, financial services, communication networks, water and sewer systems, and health services. The reliability and security of these critical infrastructure systems are paramount, as disruptions or failures in any of these areas can have far-reaching consequences on the functioning of a society. The increasing interconnectedness of these systems also emphasizes the importance of robust cybersecurity measures to safeguard against potential threats. The increase in both the frequency and severity of cyber attacks against critical infrastructure systems is indeed a growing concern. Critical infrastructure systems are often complex and interconnected, involving various components and technologies. This complexity makes it challenging to identify and mitigate vulnerabilities effectively. The geographical distribution of critical infrastructure components adds another layer of complexity. Distributed systems and remote locations can be more challenging to secure adequately. Many critical infrastructure systems rely on the Internet for efficient operation. While this enhances functionality, it also introduces additional points of vulnerability that attackers may exploit. Cyber attackers are becoming increasingly sophisticated, using advanced techniques to breach security measures.

The rise of nation-state-sponsored attacks and the availability of powerful hacking tools contribute to the severity of the threats. The growing interconnectivity of critical infrastructure systems, often referred to as the “Internet of Things (IoT)” in this context, expand the attack surface. More connected devices mean more potential entry points for cyber threats. Addressing these challenges requires a multifaceted approach, including robust cybersecurity measures, continuous monitoring, threat intelligence sharing, and collaboration between public and private sectors. The security of critical infrastructure systems: The unique complexity and structure of critical infrastructure systems introduce specific security threats not encountered in other systems. Cyber threats can exploit vulnerabilities during data transmission over networks, jeopardizing the integrity and confidentiality of critical information. Threats arise from vulnerabilities in communication protocols used in SCADA (Supervisory Control and Data Acquisition) systems, such as Modbus/TCP and DNP3, which may lack encryption, making them susceptible to unauthorized access and modification. The constant availability of critical infrastructure systems makes them potential targets for cyber attacks at any time. This continuous operation requirement increases the risk of vulnerabilities being exploited. Attacks on components like SCADA centers, remote substations, or corporate networks can have cascading effects, potentially putting the entire critical infrastructure system at risk (Alcaraz and Zeadally. 2015). The lack of encryption in certain communication protocols further exposes these systems to unauthorized access and manipulation. Given the challenges posed by structural complexity, geographical dispersion, and reliance on the Internet, securing critical infrastructure systems requires a comprehensive and adaptive cybersecurity strategy. This includes implementing robust network security measures, encrypting communication protocols, continuous monitoring, and fostering collaboration between stakeholders to address emerging threats effectively.

Reasons Arising from the Increase in Knowledge

Over the years, the accessibility of knowledge and tools has indeed lowered the barriers for launching cyber attacks. In the 1990s and early 2000s, launching attacks often required a

higher level of technical expertise, limiting the number of individuals with the capability to conduct such activities. However, with the growth of the internet, the availability of hacking tools, and the sharing of information through online forums, the landscape has changed. Today, individuals with varying levels of technical skill can access resources and tools to initiate cyber attacks. The landscape of cyber attacks has evolved significantly. In the past, initiating cyber attacks required a high level of expertise, limiting the pool of potential attackers to those with extensive knowledge in computer systems. However, with the widespread availability of attack tools, the rapid dissemination of knowledge, and the ease of identifying vulnerabilities in software and network protocols, the barrier to entry has lowered.

The emergence of “script kiddies” — individuals with limited technical knowledge who use pre-existing tools and scripts to launch attacks — exemplifies this shift. Cyber-attacks-as-a-service platforms further enable individuals without in-depth technical expertise to initiate attacks, making such tools and services accessible on the internet. The complexity of cyber attacks is increasing while the level of understanding of attackers is decreasing (Lipso. 2002; Ramirez. 2017), it implies a concerning trend. This scenario could suggest that attackers are relying more on readily available tools, techniques, or services rather than deeply understanding the underlying technology (**Figure 3**).

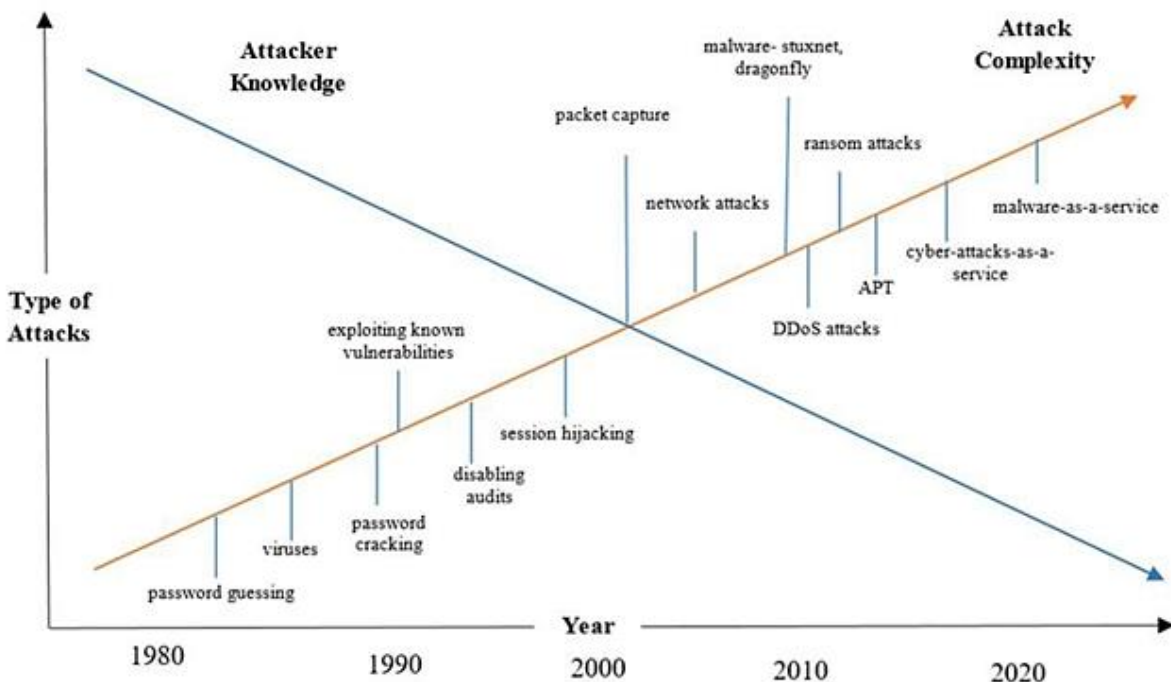


Figure 3. Technical knowledge of attackers versus attack complexity

Transferring Daily Life into the Digital Environment

The shift to the virtual environment has been a notable trend, and the COVID-19 pandemic has indeed accelerated this transition. With more people spending significant amounts of time on the internet, various aspects of social life, from socializing to business

activities, have migrated to the digital realm (Alcaraz. 2013). Virtual communication, online collaboration, and remote work have become integral parts of daily life, reshaping how individuals interact. The profound impact of virtualization on various aspects of daily life. Social media platforms like Twitter, Facebook, and Instagram play a central role in daily life, with billions of users. However, the widespread use of social media raises concerns about data privacy, security, and the potential for cyber attacks. The shift to online banking, credit cards, and virtual currencies has transformed financial transactions. However, this digital transformation introduces new vulnerabilities, leading to cyber attacks and financial losses. Remote work and virtual meetings have become commonplace, especially with the increased reliance on digital communication tools and platforms (Nurse et al. 2011). The adoption of online learning platforms has virtualized education, providing flexibility but also presenting cybersecurity challenges.

The digital realm has transformed how news is consumed, political activities are conducted, and shopping is done, with implications for information security and privacy. The mention of the virtualization of wars suggests the impact of technology on modern conflict, potentially referring to cyber warfare or the use of digital tools in traditional conflicts. Online gaming platforms and virtual environments have become integral parts of entertainment, but they are not immune to cyber threats.

The increasing prevalence of cyber attacks across these virtualized domains emphasizes the critical need for robust cybersecurity measures to safeguard personal information, financial transactions, and various online activities (Chowdhury. 2016; Abdullah et al. 2018). As technology continues to advance, addressing these challenges will remain a dynamic and ongoing process (Meszaros and Buchalceva. 2017).

The Attacks Have No Geographical Boundaries, Which Makes Detecting Attacks Challenging

Cyber attackers can operate around the clock from any location globally, exploiting the accessibility and reach of the internet. This lack of geographical boundaries makes it challenging to pinpoint and apprehend attackers. Attackers use techniques to conceal their identities and locations during cyber attacks, adding another layer of difficulty in tracking and prosecuting them. The absence of comprehensive international laws or agreements to deter cyber attacks across countries creates a permissive environment for cyber criminals. The absence of clear frameworks for punishment can embolden attackers. The inherent vulnerabilities in internet technologies and the interconnected nature of digital systems provide attackers with numerous potential entry points, making it easier to launch attacks. Addressing the global nature of cyber threats requires international cooperation, shared cybersecurity standards, and legal frameworks that transcend geographical boundaries.

The multifaceted reasons contributing to the increase in cyber attacks. Both technical and non-technical factors play significant roles in the dynamic landscape of cybersecurity. Cyber attacks constantly evolve, altering their formats and targets, impacting a wide range of computer-based systems and the internet environment. The migration of social activities to the internet amplifies the occurrence and consequences of cyber attacks, emphasizing the need for robust cybersecurity measures. Errors and vulnerabilities in software, deficiencies in network protocols, the proliferation of connected devices, and the complexity of critical systems contribute to heightened cybersecurity risks. The virtualization of social interactions, excessive use of social networks, and the increased knowledge of attackers collectively elevate security

risks in the digital realm. Careless use of the internet by users adds another layer of risk, emphasizing the importance of cybersecurity awareness and responsible online practices. Recognizing the interconnectedness of these technical and behavioral factors is crucial for developing comprehensive cybersecurity strategies that address the evolving challenges posed by cyber threats.

3. THREATS, VULNERABILITIES, EXPLOITS, AND ATTACKS

With the development of the cyber security concept, the number of attacks, types.

Of attacks, and damage they cause are also increasing. Hackers carry out attacks that target users, companies, and government institutions, causing great harm. These attacks usually include ransomware, phishing attacks, DDoS attacks, or mobile threats (Abomhara and Koien. 2015). In this section, our study extensively explores well-known cyber threats, security risks, vulnerabilities, and attacks. Our study takes a focused approach by first examining prominent cyber threats such as viruses, Trojans, worms, rootkits, and hackers. Following this, it expands the discussion to include other well-known threats like spyware, scareware, joke programs, and ransomware. Security vulnerabilities and introduces commonly used vulnerability scanning tools. The subsequent detailed analysis covers a range of common attack types, including social engineering, applications, cryptography, hijacking, computer networks, phishing, malware, bots, botnets, password attacks, and man-in-the-middle attacks. Furthermore, your inclusion of recommendations, precautions, and awareness for each type of attack enhances the practical utility of the information provided.

Threats



Figure 4. Computer Threats

Cyber threats can take various forms, such as malware, phishing, ransomware, or even direct attacks on vulnerabilities in computer systems (Abomhara and Koien. 2015; Jouini et al. 2014). It's crucial to stay vigilant and employ security measures to protect against these threats (Figure 4).

Computer viruses

Computer viruses are malicious programs designed to alter the way a computer operates without the user's permission or knowledge. They often hide within other files and can replicate themselves, spreading to other files and potentially other computers (Khan. 2012). Viruses typically become active when the infected file is executed. A computer's operation without user consent and its tendency to hide within other files. They often attempt to replicate and spread by copying themselves into other files, and their activation typically requires the user or the operating system to execute or "run" the infected file (Khan. 2022). Contrary to common belief, merely inserting an infected CD, floppy disk, or flash drive into a computer won't result in infection if there's no auto run file. The virus typically needs an active step, such as the operating system running the infected file or manual execution, for it to become active and potentially harm the system. The evolution of infectious files and the expanding range of file types that can harbor potentially harmful content. The historical focus on executable files with extensions like ".exe" has shifted as software technologies advanced. You rightly point out that even seemingly innocuous file types like ".doc" (commonly associated with text files) or ".gif" (typically recognized as image files) can now pose risks due to the integration of executable programs within them. This underscores the importance of staying vigilant and employing up-to-date security measures to protect against a broader spectrum of file-based threats.

Computer worms

Worms are indeed a type of malicious software with a more intricate structure compared to traditional viruses. Their propagation methods often involve exploiting email attachments, websites, and shared network files. One notable characteristic of worms is their ability to autonomously spread through a system, utilizing the user's data sources like email address lists, without requiring further action from the user. This self-replicating nature allows them to quickly proliferate and potentially affect a large number of users (Rajesh. Et al. 2015). Their ability to utilize users' bandwidth and network resources can indeed lead to network crashes, overload email servers, and slow down access to web resources. Your emphasis on the importance of keeping operating systems updated, performing regular updates, and using personal firewalls aligns with best practices for protecting against worm infections.

Trojan horses

These malicious programs disguise themselves as legitimate and useful software, tricking users into downloading them (Jaiswal. 2017). Trojans become active when users run the seemingly harmless program file, initiating their harmful activities. The two files in Trojan horses, with the first file opening a port on the user's computer and the second file allowing the attacker to access the system, provides insight into their modus operandi. The potential consequences, such as unauthorized access and the theft of sensitive information like passwords and credit card details, underscore the serious risks associated with Trojan infections. Trojans leaving backdoors in the system design, allowing attackers to easily install additional malware,

contribute to a comprehensive understanding of the severity of Trojan infections. The diversity of Trojan malware indeed makes it challenging to combat, emphasizing the need for a robust defense strategy. Our advice on employing an in-depth defense strategy and regularly updating all software, not just the operating system, aligns with best practices for cybersecurity. Regular software updates help patch vulnerabilities, reducing the risk of exploitation by Trojans and other forms of malware.

Rootkits

These types of malware are designed to grant unauthorized access and control over a victim's device (Bickford et al. 2010). The ability of rootkits to affect both operating systems and, in some cases, hardware and software programs underscores their potential impact on various layers of a system. The stealthy nature of rootkits, their capability to conceal their existence, and the potential consequences, such as unauthorized access, data theft, and participation in malicious activities like DDoS attacks, highlight the seriousness of these security threats. It is difficult to determine which files the rootkit actually changed, which module is loaded into the kernel, and which network service it would listen to and take action on with the appropriate command. Our mention of the difficulty in detecting rootkits and the suggested methods, like saving eigenvalues of basic commands for later comparison, provides insights into the challenges of rootkit detection.

Hackers and predators

Hackers and hunters, where hackers may sacrifice others for personal gain, provide an interesting perspective. It highlights the ethical dimension, suggesting that hackers, in this context, act with malicious intent, compromising systems for personal benefit (Patil et al. 2017). The mention of informal users attacking systems without permission aligns with the common understanding of malicious. Our emphasis on the methods employed by online hackers and predators, such as phishing scams, spam, and other techniques to compromise security, underscores the importance of vigilance for anyone using a computer connected to the internet. Employees and users posing potential threats to organizations by not adhering to safe computer practices are quite accurate. Well-trained employees serve as the initial line of defense against cyberattacks, and their lack of cybersecurity knowledge can indeed present significant risks to network security. The mention of hackers attempting direct access to computers and private information in the absence of proper protection, as well as predators disguising themselves as false identities to exploit sensitive information emphasizes the multifaceted nature of cybersecurity threats. Our recommendation for quality firewall solutions and antivirus protection aligns with best practices to safeguard against malicious activities. These measures play a crucial role in fortifying the defense against cyber threats.

Risks

The nature of technological developments, bringing both innovations and challenges, is well-founded. The shift toward distributed structures in data centers and cloud environments, while catering to the needs of the digital age, also introduces various risks to corporate and personal data (Dahbur et al. 2011). The recognition of the increasing value of data and the evolving tactics of cyber attackers, spanning activities from data theft to ransomware attacks, underlines the concept of "Cyber Risks." This term encapsulates the diverse threats that

organizations and individuals face in the digital landscape. To provide information on various types of cyber risks.

Spyware

A type of malicious software focused on collecting data without the user's consent or knowledge. The invasive nature of spyware is highlighted through actions such as recording keystrokes, tracking web pages, scanning hard disk data, and monitoring internet searches. The consequences you mention, including illegal activities such as stealing passwords, creating unwanted pop-ups and spam, as well as consuming network and system resources, illustrate the serious impact spyware can have on both privacy and the overall performance of a computer (Javaheri et al. 2018).

Spyware and viruses is insightful. Spyware is often differentiated from viruses by the fact that it is not explicitly designed for immediate and illegal harm to the user. Additionally, spyware is typically installed on a computer with the user's consent or as a result of their actions, although this consent may not always be well-informed. Our suggested security measures, such as avoiding downloads from untrusted sources, refraining from connecting untrusted devices, and exercising caution with unknown emails, align with best practices for preventing spyware infections. Being vigilant in these areas can contribute significantly to enhancing cybersecurity.

Scareware

A type of malicious software commonly referred to as "Rogueware" (Chen et al.2019). This category often involves deceptive tactics, where users are frightened with unrealistic scenarios to encourage them to purchase the supposed protective software. However, the installed software itself can be malicious, leading to further harm rather than protection. The malicious software, after being installed under the guise of cleaning the system, may actually infect the device with a virus. The consequences involve intercepted data and restricted access to the computer's owner.

The deceptive tactics employed by scareware. The primary objective is to generate revenue for its owner by instilling fear in users. The scenario you describe, where a pop-up falsely claims threats on the computer and then suggests purchasing the scareware software for protection, underscores the manipulative nature of this type of malware. Highlighting that scareware often fails to deliver on its promises and primarily aims to extract payment from users is an important caution. Your advice on finding and removing scareware through the computer's Program, Add/Remove menu is practical and can help users take corrective action.

Joke programs

Joke programs provide a helpful distinction between them and more malicious types of software. Joke programs are typically created for entertainment or to annoy users without a primary malicious purpose (Wadhwa and Arora. 2017). The intent is often to make fun of computer users rather than to harm them. However, there can be unintended consequences, especially if users panic and engage in activities like drive formatting or file deletion, potentially causing damage to their data. It's important to note that while joke programs may be disruptive or annoying, they generally don't cause lasting harm to other programs, systems, or data. The occasional temporary deactivation of devices, like a mouse or printer, is mentioned

as a possible effect. The computer returning to its original state after a joke program ends or the user restarts it provides a crucial context. It emphasizes that, unlike more malicious software, the impact of joke programs is typically temporary and reversible. The acknowledgment that joke programs, while seemingly innocent, can sometimes incur costs for institutions or organizations is an important point. This is particularly relevant in environments where disruptions, even if temporary, can have operational or financial implications.

Ransomware

The severe impact and modus operandi of ransomware. It is indeed one of the most prominent and disruptive types of cyber risks. The ability of ransomware to encrypt essential files, documents, applications, and entire networks from a single computer is a major concern. The mention of victims being asked to pay a ransom in cryptocurrency, with the promise of data restoration upon payment, underscores the extortionate nature of ransomware attacks (AL-Rimy et al. 2018). It's a tactic that cybercriminals use to exploit the urgency and importance of the encrypted data to the victims. The uncertainty of whether criminals retain copies of data even after the ransom is paid highlights a critical aspect of dealing with ransomware attacks. While victims may regain access to their files after paying the ransom, the lingering threat of potential misuse in fraud or phishing attempts remains. The caution about deceptive tactics, such as using fake attachments or links disguised as ordinary documents from reputable sources, is valuable advice. Ransomware often infiltrates systems through seemingly innocuous files, exploiting the trust associated with familiar document types like order confirmations or invoices.

Hacking tools

The deliberate modification of computer software or hardware outside architectural boundaries and design is accurate. Additionally, the clarification that hacking tools are programs designed to assist hackers in their activities is insightful. The acknowledgment that hacking tools can be used both for malicious purposes and proactively to protect networks or computers adds nuance to the discussion. The example of Hacktool: Win32/Keygen highlights a specific hacking tool that generates counterfeit activation keys or licenses for various software, illustrating the diverse capabilities of such tools (Pham et al. 2010). The device itself may not be malicious, the presence of Hacktool: Win32/Keygen often bundled with malware is crucial information. Users who have installed this tool, whether knowingly or without their permission, are at risk of having an infected computer. The mention of hacking tools being commonly used to gain unauthorized access and introduce more malicious elements like worms, viruses, and Trojans underscores the serious consequences associated with their use. The potential outcomes, such as loss of data, hijacking personal accounts, and theft of identity and savings, emphasize the significant risks posed by hacking tools.

Remote access

Remote access is functionality and the associated risks. While remote access is a valuable application for managing computers over the internet from any location, your caution about the risks is well-placed. The recognition that many individuals, institutions, and organizations may not adequately protect remote connection services, creating opportunities for inappropriate access, is an important consideration (Clausen et al. 2020). The potential consequences of

remote attacks, such as the illegal viewing or stealing of data, introducing malware, and causing damage to targeted systems, underline the seriousness of these risks.our mention of various attack vectors facilitated via remote access, including email phishing, compromise of third-party providers, insider threats, social engineering, and the use of vulnerable applications, provides a comprehensive overview of the potential vulnerabilities associated with remote connections. Protecting against remote access attacks are well-founded and align with established best practices in cybersecurity: This is a critical point. Using up-to-date and secure VPN protocols helps mitigate vulnerabilities associated with outdated or less secure options. Enforcing multifactor authentication for each VPN account adds an extra layer of security, making it more challenging for unauthorized users to gain access. Applying at least two layers of security before allowing remote device management is a prudent approach. This could involve a combination of secure authentication methods, access controls, and encryption. Keeping antivirus and antimalware programs installed and up-to-date is essential for ongoing protection against various threats, including those that may attempt to exploit remote access. These measures collectively contribute to a robust defense against potential remote access attacks.

Vulnerabilities

Vulnerabilities refer to weaknesses or flaws in software, hardware, or systems that could be exploited by attackers to compromise their security. These weaknesses could exist in various components, such as code, configurations, or design, making the system susceptible to unauthorized access, data breaches, or other security threats (Humayun et al. 2020). Addressing vulnerabilities is a critical aspect of cybersecurity, involving practices like regular software updates, patch management, and implementing security best practices to reduce the risk of exploitation.

Software vulnerabilities

Software vulnerabilities arise when applications contain errors or bugs, providing attackers with opportunities to exploit these weaknesses and compromise the system. Buffer Overflow occurs when a program writes more data to a block of memory, or buffer, than it was allocated for. This overflow can lead to unpredictable behavior, crashes, or, in more malicious scenarios, the execution of arbitrary code. Race Conditions occur when the behavior of a program depends on the relative timing of events, and an attacker manipulates this timing to disrupt or compromise the system. Race conditions can lead to unexpected outcomes and potentially be exploited for malicious purposes. Addressing and patching these vulnerabilities through regular software updates and implementing secure coding practices are crucial steps in maintaining a resilient and secure computing environment.

Firewall vulnerabilities

Firewalls, whether implemented as software or hardware safeguards, play a crucial role in protecting networks from various attacks. Firewall vulnerability occurs when there is a mistake, deficiency, or faulty assumption made during the design, implementation, or configuration of the firewall. These vulnerabilities can be exploited by attackers to launch attacks against the trusted network that the firewall is intended to protect. Addressing firewall vulnerabilities is essential for maintaining the effectiveness of these security measures. Regular

updates, proper configuration, and adherence to security best practices are key elements in reducing the risk of firewall vulnerabilities and enhancing overall network security.

TCP/IP vulnerabilities

TCP/IP vulnerabilities can manifest across various layers of a network, and these vulnerabilities may result from the absence of desired features in an unsecured network. ARP Attacks (Address Resolution Protocol Attacks) is a protocol used to map IP addresses to physical MAC addresses. ARP attacks involve manipulating or poisoning the ARP cache, leading to misdirection of network traffic. Fragmentation Attacks exploit vulnerabilities in the way network packets are fragmented and reassembled. By manipulating the fragmentation process, attackers may disrupt communication or even execute malicious actions. Addressing TCP/IP vulnerabilities involves implementing security measures, such as encryption, secure configurations, and monitoring, to mitigate the risks associated with potential attacks.

Wireless network vulnerabilities

To wireless local area networks (LANs) and the vulnerabilities they may face. Wireless LANs, like their wired counterparts, can be susceptible to protocol-based attacks. Additionally, insecure wireless access points pose a potential threat by serving as entry points for unauthorized access. Service Set Identifier (SSID) Issues is a unique identifier for a wireless network. Problems with SSID settings, such as using default or easily guessable values, can expose the network to security risks. Wired Equivalent Privacy (WEP) Issues was an early security protocol for wireless networks. However, it is now considered insecure due to vulnerabilities that can be exploited by attackers. Securing wireless LANs involves implementing robust security protocols like WPA3, regularly updating access point configurations, and employing strong encryption methods to mitigate potential risks.

Operating system vulnerabilities

The critical role that operating systems play in the overall security of a computing environment. Security vulnerabilities in operating systems, whether it's Windows, macOS, Unix, or others, can have significant implications for the security of the entire system. The security of applications running on these operating systems is inherently tied to the security measures implemented at the OS level.

Web server

Web server vulnerabilities and the potential attacks that can result from them. Web server vulnerabilities indeed stem from design and engineering errors or misapplication, creating opportunities for malicious actors to exploit weaknesses. Sniffing Attacks involve the unauthorized interception and monitoring of network traffic. In the context of web servers, sniffing attacks could compromise sensitive data being transmitted between clients and the server. Spoofing attacks involve the manipulation of identity or information to deceive users or systems. In the context of web servers, spoofing attacks could lead to users being directed to fraudulent websites or attackers masquerading as legitimate entities. Addressing web server vulnerabilities requires proactive measures such as regular security audits, prompt application of patches, and adherence to secure coding practices to minimize the risk of exploitation.

Vulnerabilities scanning tools

Vulnerability scanning tools and their role in detecting security vulnerabilities in devices and software within systems (Tundis et al. 2018). The analogy to antivirus software using virus definition databases to identify virus signatures helps illustrate the concept. Vulnerability scanners rely on databases containing information about specific vulnerabilities. These databases are often compiled from reputable security testing labs or known sources dedicated to identifying vulnerabilities in software and hardware. The effectiveness of a vulnerability scanning tool depends on the comprehensiveness and accuracy of the vulnerability database it uses. A robust and up-to-date database is crucial for identifying potential security risks. When selecting a vulnerability scanning tool, factors such as the variety of devices it can scan, scanning methods employed, and the quality of warning/reporting features should be taken into account.

Nets-parker

Netsparker highlights its accuracy and automation in scanning for vulnerabilities in web applications and APIs. The emphasis on Netsparker's capability to heavily confirm defined vulnerabilities and distinguish them from false positives is a notable feature. This characteristic can significantly reduce the need for manual confirmation, streamlining the scanning process. The availability of both Windows software and an online service version provides users with flexibility in choosing the deployment method that best suits their needs (Joshi and Singh. 2016).

Acunetix

Acunetix highlights its features as a fully automated web vulnerability scanner with a comprehensive capability to detect over 4500 web application vulnerabilities, including SQL Injections and XSS (Cross-Site Scripting). The mention of providing detailed reports adds value to its usability for users looking to understand and address identified vulnerabilities. Acunetix supports single-page applications with HTML5 and JavaScript, showcasing its adaptability to inspect complex and authenticated applications. Sophisticated Vulnerability Management tool incorporates sophisticated vulnerability management functionalities, allowing for the ranking of risks based on information gathered from a unified and consolidated perspective. Acunetix goes beyond scanning by combining its findings with additional tools and platforms, providing a more holistic approach to addressing vulnerabilities (Joshi and Singh. 2016).

Intruder

The Intruder provides valuable insights into its features and capabilities as an advanced vulnerability scanner (Wang and Yang. 2017). Proactive Scans for Newly Released Vulnerabilities intruder stands out for its proactive approach, conducting scans for newly released vulnerabilities to enhance security. Extensive Security Checks With over 10,000 security checks, including coverage for notable vulnerabilities like Heartbleed and SQL Injection, The Intruder demonstrates a robust scanning capability. Integration with collaboration tools such as Slack and Jira enables development teams to stay updated in real-time when new security issues arise. AWS Integration with AWS facilitates the synchronization of IP addresses for scanning, streamlining the scanning process for cloud-based environments. Preferred Choice for Startups and Midsize Businesses: The Intruder is positioned as a preferred

choice for startups and midsize businesses, offering simplified vulnerability management tailored for small teams.

Solar-Winds

The Solar-Winds Network Configuration Manager sheds light on its network vulnerability detection capabilities and its broader functionality in monitoring and safeguarding network configurations. Network Vulnerability Detection tool incorporates capabilities for detecting vulnerabilities within the network, providing a proactive approach to security. Solar-Winds Network Configuration Manager is designed to monitor, manage, and safeguard network configurations, emphasizing a comprehensive approach to network security. Notification System for Configuration Changestool includes a notification system that alerts administrators when configuration changes occur, helping to maintain awareness of alterations to the network. Continuous checks are performed to identify configurations that may render a device incompatible, contributing to a proactive stance in preventing potential issues. The software enables configuration backup, facilitating the monitoring of changes and serving as a valuable resource for restoring configurations when needed. Users can obtain detailed information about configuration modifications, including the login ID associated with them (Peisert et al. 2021). This adds transparency and accountability to configuration management.

Apptrana

The web application vulnerability scanner, emphasizing its features and practical applications. Automation of Detection and Reporting tool automates the detection and reporting of security vulnerabilities in web applications, streamlining the process for users. Wide User Base with over 1100 customers across 25+ countries worldwide, the tool has established itself as a widely used solution in the field of web application security. Users can utilize a proof-of-concept request to validate reported vulnerabilities and eliminate false positives, enhancing the accuracy of the scanning results. The same dashboard displays additional manual penetration tests and reports, providing a unified view of automated and manual security assessments. The tool features a pause and resume function, offering flexibility in managing scanning processes according to user preferences. Users have the option for integration with Indus face WAF, providing immediate virtual patching with zero false positives, enhancing the overall security posture (Paez and Michel. 2021).

Open VAS

Open VAS provides a clear understanding of its key attributes as an open-source tool for centralized vulnerability scanning and management. OpenVAS is an open-source tool, making it accessible at no cost to users. The use of the GNU General Public License (GPL) further emphasizes its open and community-driven nature. Compatibility software is compatible with a range of operating systems, enhancing its versatility and adaptability to different environments. OpenVAS features a regularly updated scan engine that incorporates network vulnerability tests. This ensures that the tool remains current and effective in identifying security-related issues. OpenVAS serves as a comprehensive vulnerability assessment tool, designed to identify security-related issues on servers and other network devices. This highlights its role in providing a thorough evaluation of the security posture. The combination

of its open-source nature, compatibility, and comprehensive scanning capabilities makes OpenVAS a valuable tool in the realm of vulnerability assessment (Rahalkar. 2019).

The nexpose community

The Nexpose vulnerability scanner, developed by Rapid7, provides valuable insights into its features and capabilities. Nexpose serves as a tool for performing network checks and vulnerability scans, contributing to the ongoing monitoring of vulnerability exposure. Real-time Vulnerability Exposure Monitoring: The tool is employed for real-time monitoring of vulnerability exposure, allowing organizations to stay informed about emerging threats using new data. Advanced Risk Ranking unlike some other vulnerability scanners that use high, medium, or low risk classifications, Nexpose employs a more nuanced approach, considering factors such as the specific malware kit used and the benefits derived from vulnerability. This allows for a more sophisticated prioritization of security vulnerabilities. Automatic Identification and Scanning of New Devices. Nexpose has the capability to automatically identify and scan new devices when they connect to the network, ensuring that the assessment of vulnerabilities remains up-to-date. Integration with Metasploit Framework with the Metasploit framework enhances Nexpose's capabilities by providing additional tools and resources for addressing vulnerabilities. Overall, Nexpose appears to offer a comprehensive and dynamic approach to vulnerability assessment, taking into account real-time monitoring and advanced risk considerations (Roldan-Molina et al. 2017).

Nikto

Nikto role as an open-source web scanner focused on evaluating potential web server vulnerabilities. Nikto is utilized for evaluating potential vulnerabilities and issues within web servers, emphasizing its role in web security assessment. Thorough Testtool conducts thorough tests on web servers to identify hazardous programs or files, ensuring a comprehensive examination of potential security risks. Nikto verifies that the server version is current and identifies any issues that may impact server operation. This aspect contributes to maintaining the security and stability of web servers. Nikto scans various protocols, including HTTP, HTTPS, and HTTPD, showcasing its versatility in assessing web server security across different communication protocols. Scanning Multiple Server Ports tool supports the scanning of multiple server ports, allowing for a broader assessment of the server's security posture. Overall, Nikto appears to be a versatile and widely favored tool for conducting thorough assessments of web server vulnerabilities, making it a valuable asset in the field of web security (Karangle et al. 2019).

Microsoft baseline security analyzer (MBSA)

The Microsoft Baseline Security Analyzer (MBSA), emphasizing its role in detecting security status, scanning for vulnerabilities, and monitoring operating systems in a network. MBSA is designed to detect the appropriate security status by aligning with Microsoft security recommendations. This ensures that systems adhere to recommended security practices. Vulnerability Scanning tool scans for security vulnerabilities on computers running Windows operating systems, contributing to the identification and remediation of potential security risks. MBSA can monitor all operating systems in the network, allowing administrators to assess the security posture across multiple systems. To perform network-wide monitoring, the

administrator's user and password on the computer where MBSA is installed must match those of the computers being scanned. This authentication ensures the tool's access to relevant information. MBSA includes the capability to perform HOTFIX checks for IIS (Internet Information Services) and SQL servers, addressing additional security considerations beyond the operating system. Overall, MBSA appears to be a versatile tool that provides administrators with insights into security vulnerabilities, facilitates network-wide monitoring, and checks for relevant updates in specific server components (Pattanavichai et al. 2017).

Wireshark

Wireshark, highlighting its significance as a globally recognized network protocol analyzer. Wireshark is employed in diverse settings, ranging from educational institutions to government agencies and businesses. This versatility speaks to its applicability across various domains. Detailed Network Examination tool is utilized for conducting detailed examinations of networks, enabling users to gain insights into network traffic and diagnose potential issues. Wireshark possesses the unique capability to detect problems in real time, allowing users to identify and address issues as they occur. In addition to real-time analysis, Wireshark can conduct analyses offline. This feature provides flexibility in reviewing captured network data at a later time. Wireshark is compatible with multiple platforms, including Windows, Linux, Mac, and Solaris. This broad compatibility ensures accessibility for users across different operating systems. Detailed Protocol Exploration tool can explore numerous protocols in great detail, offering a comprehensive view of the various protocols operating within a network. Wireshark is widely considered the most robust tool in the arsenal of security professionals, underscoring its importance in the field of network security. Overall, Wireshark's widespread usage, detailed analysis capabilities, and compatibility across platforms contribute to its reputation as a powerful network protocol analyzer (Ndatinya et al. 2015).

Attacks

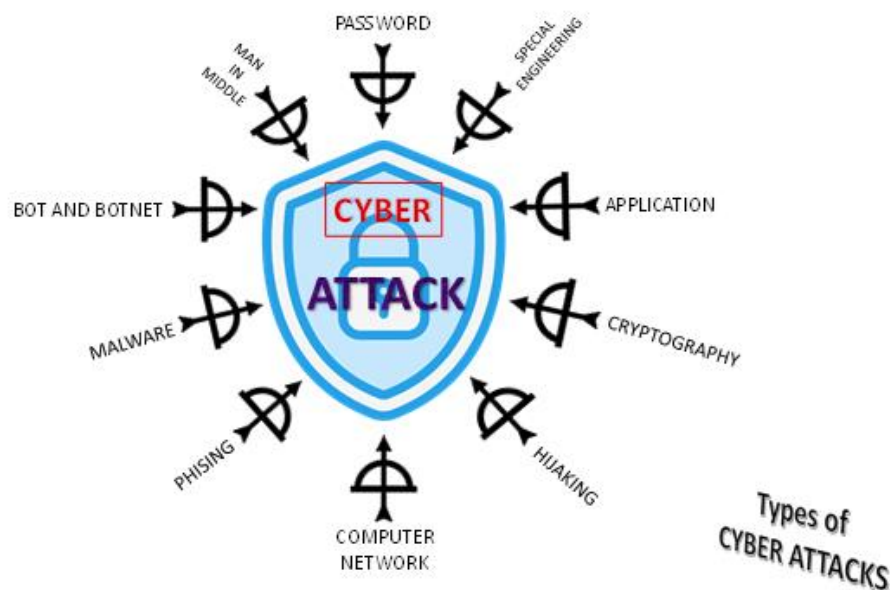


Figure 5. Types of cyber attacks

The development of technology has brought about significant advancements and improved accessibility to information. However, this increased ease of access has also given rise to challenges in ensuring information security. The prevalence of information systems in various aspects of life has made the concept of cyber attacks more widespread. Cyber attacks have evolved into a pervasive threat affecting diverse domains, including daily life, government institutions, the economy, commerce, banks, and even hospitals. The interconnected nature of digital systems and the reliance on technology in various sectors have made cybersecurity a critical concern. As technology continues to advance, the need for robust cybersecurity measures becomes increasingly crucial to safeguard sensitive information, maintain privacy, and ensure the integrity and availability of digital systems. Common type's cyber attacks describe as **Figure 5**.

Social engineering attacks

Social engineering is a technique that involves manipulating individuals to gain unauthorized access to information or data. Unlike traditional cyber attacks that exploit technological vulnerabilities, social engineering exploits human vulnerabilities by tricking individuals into divulging sensitive information or granting access to data networks. This form of manipulation can take various forms, including phishing emails, pretexting, baiting, or even impersonation. The ultimate goal of social engineering is to bypass the security measures implemented by individuals or organizations, relying on human factors such as trust, curiosity, or fear. Social engineering attacks can target anyone, but certain groups may be more vulnerable due to factors like limited technical knowledge, social isolation, or impulsive behavior. The elderly, in particular, are often targeted, as they may have less familiarity with modern technology and can be more trusting. Avoid sharing personal or private information unless absolutely necessary. Be cautious about what details you disclose online or over the phone. Question unexpected communications, especially those requesting sensitive information. Verify the identity of the person or organization contacting you before sharing any details (Krombholz et al. 2015). Check URLs and Addresses be vigilant about the websites you visit and the links you click. Verify the legitimacy of URLs and email addresses to avoid falling victim to phishing attempts. Avoid Untrustworthy Sources be skeptical of unsolicited messages, emails, or phone calls, especially if they seem suspicious. Avoid engaging with untrustworthy sources to minimize the risk of social engineering attacks. By adopting these practices, individuals can significantly reduce the likelihood of falling victim to social engineering tactics and enhance their overall cybersecurity resilience. Education and awareness play key roles in empowering people to recognize and respond appropriately to potential threats.

Application attacks

Application attacks and the common methods employed by cybercriminals. Application attacks involve unauthorized access to domains, and attackers typically begin by scrutinizing the application layer for vulnerabilities present in the code. Application attacks concentrate on the application layer, where cybercriminals search for vulnerabilities within the code. Programming Language Targets while certain programming languages may be targeted more frequently, applications written in various languages are susceptible to attacks. Vulnerabilities in Code both proprietary code and open-source frameworks/libraries may contain vulnerabilities that attackers can exploit. Exploitation in Production: application vulnerabilities

create opportunities for cybercriminals to exploit applications, particularly in production environments. The diversity of methods employed by cybercriminals when exploiting application vulnerabilities. Code Vulnerabilities: Both proprietary code and open-source frameworks/libraries can contain vulnerabilities that cybercriminals exploit. These vulnerabilities may result from coding errors or oversights. Exploiting vulnerabilities related to legacy certificates indicates that outdated or improperly configured security certificates can be targeted. This can lead to unauthorized access or other security breaches. Vulnerabilities due to a lack of authentication suggest that cybercriminals may exploit weaknesses in the authentication mechanisms of applications. This can involve bypassing login procedures or taking advantage of insufficiently protected authentication processes. Addressing these vulnerabilities requires a comprehensive approach to application security, including regular code audits, updates to frameworks and libraries, proper certificate management, and robust authentication mechanisms. Cybersecurity efforts should be ongoing to adapt to evolving threats and protect against various attack vectors (Al-Khurafi and Al-Ahmad. 2015).

Cryptography attacks

Cryptography, encryption, and the potential vulnerabilities through cryptographic attacks. Cryptography ensures the secure storage and transmission of data by converting it into a specific format, making it accessible only to intended recipients. Encryption and Decryption process involves encryption (converting plain text to cipher text) and decryption (converting cipher text back to plain text). Cryptographic Attacks occur when an adversary attempts to compromise a cryptosystem by identifying weaknesses in the code, cipher, cryptographic protocol, or key management. Passive vs. Active Attacks unauthorized access to information without disrupting communication. Active Attacks modifying or initiating an unauthorized transmission of information, compromising data integrity. It's crucial to implement strong cryptographic practices, regularly update cryptographic protocols, and manage cryptographic keys securely to defend against both passive and active attacks (Pavlenko et al. 2019). As technology evolves, staying vigilant against emerging cryptographic vulnerabilities is essential for maintaining the confidentiality and integrity of sensitive information.

Hijacking attacks

Hijacking attacks in the realm of network security. Hijacking attacks involve the unauthorized takeover of computer systems, software programs, and network communications. Many cyber attacks leverage hijacking techniques as part of their strategy. It's highlighted that hacking, in general, is often illegal and carries serious consequences for both the attacker and the victim (Vimal et al. 2017). The analogy is drawn between hijacking attacks in the digital realm and physical scenarios like airplane hijacking or taking control of an armored transport vehicle. There are various types of hijack attacks, each with its own methodology and impact. Understanding and mitigating hijacking attacks involve implementing robust security measures, including intrusion detection systems, secure authentication practices, and regular security assessments to identify and address potential vulnerabilities in network communication. Cybersecurity awareness and proactive defense are crucial components in safeguarding against hijacking threats. Several types of hijacking attacks commonly encountered in the digital realm and they are: Browser Hijacking, Session Hijacking, Domain Hijacking, Clipboard Hijacking, DNS Hijacking, IP Hijacking, and Page Hijacking.

Each type of hijacking attack poses unique threats to cybersecurity, and preventive measures should be implemented to mitigate these risks.

Computer network attacks (CNAs)

CNAs involve the intentional manipulation, corruption, rejection, or destruction of information within computers and computer networks. The primary objective is to gain control of the targeted computer or network. CNAs can result in various consequences, including shutting down systems, modifying data, and misusing resources for activities like botnets (Pawar and Anuradha. 2015). The ultimate goal is to compromise the integrity or availability of the targeted system. CNAs rely on manipulating the data stream to execute attacks. An example is sending a code or instruction to a central processing unit, causing a short circuit in the computer's power supply. To perform a CNA, attacker's first gain unauthorized access to the network. They then conduct reconnaissance (scouting) to understand the network configuration and identify optimal ways to carry out malicious activities. Understanding the network allows attackers to plan and execute their attacks effectively, exploiting vulnerabilities and weaknesses discovered during the scouting. Placental are a significant cybersecurity concern, and defending against them requires robust security measures, including access controls, encryption, intrusion detection/prevention systems, regular network monitoring, and employee training on cybersecurity best practices. Additionally, staying informed about evolving threats and applying timely security updates are crucial components of a comprehensive defense strategy.

Phishing attacks

Phishing is described as a type of online fraud that specifically targets consumers. The attackers send emails that falsely appear to come from a trusted or known source. Scammers often impersonate well-known entities such as Internet service providers, banks, companies, or government agencies. This impersonation is a crucial element of phishing attacks. The primary goal of phishing attacks is to trick individuals into divulging sensitive and confidential information. This can include usernames, passwords, credit card details, and network credentials. Phishing heavily relies on social engineering techniques (Basit et al. 2020). Attackers use psychological manipulation to exploit human behavior and persuade victims to click on harmful links or open malicious attachments. Phishing attempts can occur through various communication channels, including email and phone calls. Attackers may use a combination of these methods to increase the likelihood of success. A common tactic in phishing is to prompt victims to click on harmful links or open malicious attachments. These actions may lead to the installation of malware or the redirection to fraudulent websites designed to collect sensitive information. Emphasizing the importance of user attention and providing comprehensive education are fundamental in combating phishing attacks. Educated users are more likely to recognize and avoid phishing attempts. Deploying email monitoring systems helps filter out many common phishing attempts before they reach end-users. These systems employ various techniques, including pattern recognition and threat intelligence, to identify and block malicious emails. Conducting regular training sessions for employees is key. These sessions should cover recognizing phishing attempts, understanding common tactics employed by attackers, and reinforcing the importance of not clicking on suspicious links or opening unexpected attachments. Users should exercise caution when encountering unexpected

pop-ups, especially on websites. Legitimate websites typically do not use intrusive pop-ups, and users should avoid interacting with them. Verifying the website's security is crucial. Users should check for "HTTPS" in the URL, indicating a secure connection, and look for a closed lock icon next to the address bar. This helps ensure that sensitive information is transmitted securely. By combining these measures, organizations and individuals can significantly enhance their defenses against phishing attacks. Ongoing awareness, training, and technological solutions work together to create a more resilient cybersecurity posture.

Malware attacks

Malware encompasses a range of harmful programs designed to compromise the integrity, confidentiality, or availability of computer systems and data. Viruses that attach themselves to other files and replicate when the infected file is executed. Worms that spread across networks without the need for user interaction. Trojans malware disguised as legitimate software to trick users into installing it, often leading to unauthorized access. Rootkits software that provides unauthorized access while hiding its presence on a system. Code Variability malware can be written in various programming or scripting languages. File Types can be carried in different file types, making detection challenging. Spying and Information Gathering malware often operates silently, collecting information over extended periods without detection. Espionage gathering sensitive information without alerting the victim. Sabotage causing damage to the system or disrupting its normal operation. Extortion forcing victims to pay money or provide something of value to regain control or prevent further harm. For examples Stuxnet known for targeting industrial systems and causing physical damage. Ransomware encrypts files and demands payment for their release. Spyware collects user information without their knowledge. Combatting malware requires a multi-faceted approach, including up-to-date antivirus software, regular system updates, user education on safe computing practices, and network security measures (Kramer and Bradfield. 2010).

Bot and botnets

The word "botnet" is derived from "robot" and "network." It refers to a network of computers or devices that are infected with malicious bots, working together under the control of a single entity (Eslahi et al. 2012). Botnets are often used for coordinated and large-scale cyber attacks, leveraging the combined computing power of the infected devices to carry out various malicious activities. Botnets indeed comprise software agent programs, each remotely controlled, and they operate collectively to accomplish various tasks. Unfortunately, one common use is to send spam emails. The concerning aspect is that these networks of compromised computers, often referred to as zombie computers, can be substantial in scale, numbering in the thousands or even hundreds of thousands, all under the control of the attacker.

Password attacks

Password attacks are a common and effective method in cyber attacks. Individuals and institutions can be vulnerable if they use easily guessable passwords, especially when personal details are shared openly on social media (Raza et al. 2012). Cyber attackers often leverage such personal information to increase the success rate of password attacks, emphasizing the importance of using strong, unique passwords and being cautious about sharing sensitive information online.

Man-in-the-middle attack

The man-in-the-middle attack remains a persistent threat in cybersecurity. By secretly intercepting communication between two parties, attackers can read, modify, or even impersonate the participants. Free Wi-Fi areas are particularly vulnerable, as unencrypted packets can be easily accessed (Conti et al. 2016). Attackers redirect network traffic through their connection, enabling them to seize sensitive information like personal data or passwords. This emphasizes the importance of using secure connections, especially in public Wi-Fi environments, and employing encryption to protect data in transit. Security awareness and responsible online behavior are indeed fundamental. Utilizing licensed software, employing antivirus and antispyware tools, maintaining personal firewalls, and adopting strong passwords contribute significantly to protecting computer-based systems. Avoiding downloads from unknown sources, being cautious with email attachments, and steering clear of suspicious websites are prudent practices. Incorporating virtual machines adds an extra layer of security. In the ever-evolving landscape of cyber threats, a proactive and informed approach remains the key to safeguarding computer systems and personal information.

4. NETWORK SECURITY

In the section on “Network Security,” our paper likely addresses strategies, technologies, and measures employed to safeguard computer networks from cyber threats. This could encompass topics such as firewalls, intrusion detection systems, encryption, and other protective measures designed to ensure the integrity and security of network communications. Network security is closely tied to the rules and structures governing data exchange among connected devices, and the OSI model is a foundational framework for understanding network protocols. The Open Systems Interconnection (OSI) model consists of seven layers, each serving a specific function in network communication. These layers, from the physical transmission of data to the user interface, include the Physical, Data Link, Network, Transport, Session, Presentation, and Application layers. Understanding the OSI model aids in designing and implementing secure network protocols. Security measures can be applied at different layers to protect against various threats, ensuring the confidentiality, integrity, and availability of data during transmission across networks.

The OSI model

The OSI model, established by the International Organization for Standardization (ISO), serves as a crucial framework for comprehending network protocols. By dividing the communication process into seven independent layers, it provides a structured approach to network design and functionality. Each layer has its specific tasks and operates independently, contributing to the modularity and flexibility of the overall communication process. This modular design allows for easier troubleshooting, maintenance, and development of network protocols. The OSI model’s structured approach with its seven independent layers indeed facilitates a more organized understanding of network protocols (**Figure 6**). The modularity and independence of each layer not only contribute to flexibility in network design but also ease troubleshooting, maintenance, and the development of protocols. We now classify and explain

each layer and its functionalities with a top-down (from Layer 7 to Layer 1) approach (Zimmermann. 1980).

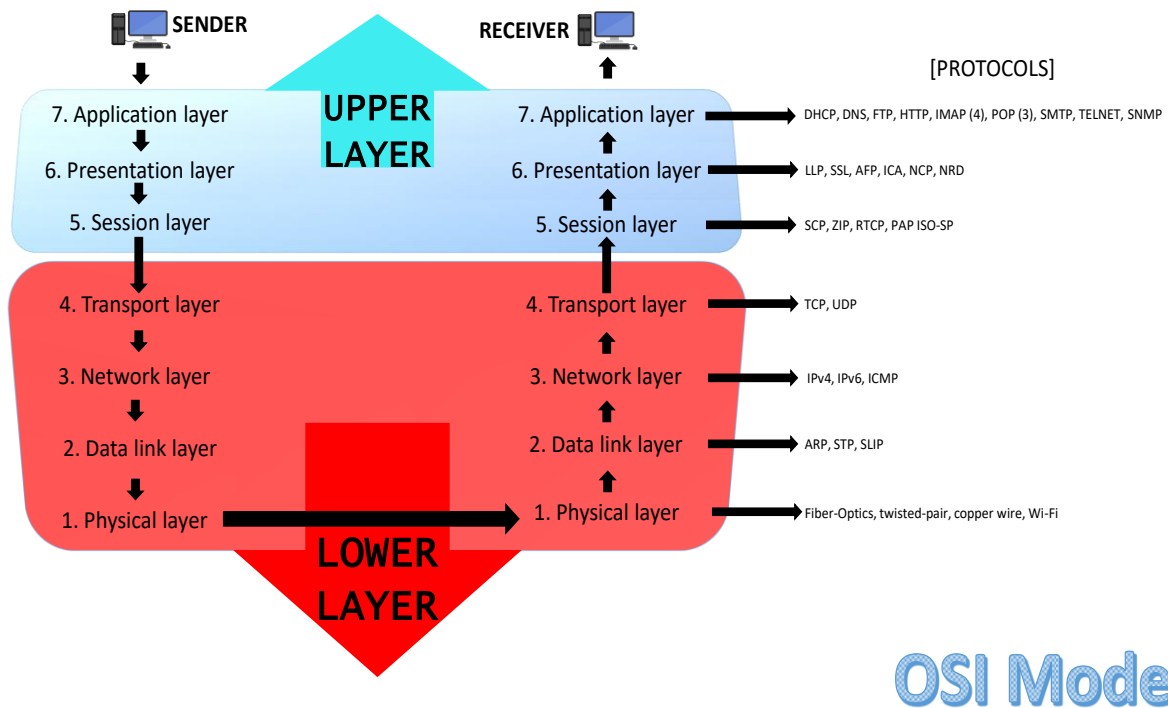


Figure 6. The OSI model

Application layer

The application layer in the OSI model is indeed concerned with how end-users communicate by defining message types and the rules for sending and responding to messages. It plays a crucial role in network protocols and applications. The mentioned application layer protocols like DHCP, DNS, FTP, HTTP, IMAP, POP, SMTP, TELNET, and SNMP are vital for various network functions, such as dynamic host configuration, domain name resolution, file transfer, web browsing, email communication, remote login, and network management (Kurose and Rose. 2012). Each protocol serves specific purposes within the application layer, contributing to the overall functionality of network communication.

❖ Application layer protocols

DHCP (Dynamic Host Configuration Protocol) and its importance is accurate. DHCP indeed plays a crucial role in automatically configuring and distributing IP addresses and other network parameters, especially in scenarios where manual configuration becomes impractical due to the increasing number of connected devices (Nastase. 2017). The DORA process—Discovery, Offer, Request, and Acknowledgment—efficiently outlines how DHCP operates in dynamically assigning IP addresses to clients (Syed et al. 2021). The growing importance of DHCP security (Raza et al. 2012). As the number of connected devices increases, ensuring the security of the DHCP process becomes critical. The lack of authentication mechanisms can

make DHCP vulnerable to attacks (Droms. 1997; Alexander and Droms. 1997), such as IP address starvation attacks or the use of rogue DHCP servers to provide incorrect configuration information (Dinu et al. 2017). Implementing DHCP security measures becomes essential to safeguard the integrity and reliability of network configurations (Alexander and Droms. 1997; Droms. 1997; Droms and Arbaugh. 2001; Tripathi and Hubballi. 2016; Younes. 2016; Dinu et al. 2017; Sayed et al. 2021).

➤ **Domain name system (DNS)**

The Domain Name System (DNS) operations are accurate and well-explained. DNS indeed plays a vital role in translating human-readable domain names into computer-friendly IP addresses, facilitating the identification and location of devices on the Internet (Ruiz-Sanchez et al. 2001). DNS resolution occurs, starting from a client's request for a domain name and involving DNS resolvers, root name servers, Top-Level Domain (TLD) servers, and domain name servers. This process is fundamental for the proper functioning of the internet, ensuring that users can access websites and other resources using user-friendly domain names rather than numerical IP addresses (Lyu et al. 2022).

➤ **File transfer protocol (FTP)**

The File Transfer Protocol (FTP) is accurate. Indeed, FTP serves as a fundamental protocol for transferring files between connected computer systems, even when they are heterogeneous in nature (Gien. 1978). However, as you rightly point out, its objectives extend beyond simple file transfers. FTP facilitates not only the sharing of files, programs, and data but also provides a means to access and utilize remote computers. Additionally, it acts as a shield, managing variations and efficiently transferring data among hosts. The versatility of FTP makes it a valuable tool in various scenarios, contributing to seamless data exchange and remote access in computer networks (Postel and Reynolds. 1985).

➤ **Hypertext transfer protocol (HTTP)**

HTTP (Hypertext Transfer Protocol) and its role in the World Wide Web is accurate. HTTP is indeed the foundation of the web, responsible for loading web pages through hypertext links. Operating in the application layer of the OSI model, HTTP runs on top of other layers and facilitates the exchange of information among connected computers. Clients, typically web browsers, make requests for various types of content such as HTML files, videos, scripts, photos, and text from a server. The server then responds by providing the requested data to the client, enabling the rendering of web pages. This client-server interaction forms the basis of how information is exchanged on the World Wide Web, highlighting the importance of HTTP in web communication (Kurose and Ross. 2012; Koksai and Tekinerdogan. 2017).

➤ **Email protocols**

The Internet Message Access Protocol (IMAP), Post Office Protocol (POP), and Simple Mail Transfer Protocol (SMTP) is comprehensive and accurate. SMTP (Simple Mail Transfer Protocol) primarily used for sending emails, SMTP is responsible for pushing emails from the sender's client to the recipient's server. It plays a crucial role in initiating the email delivery process. POP3 (Post Office Protocol version 3) typically used to retrieve emails from a server, POP3 stores emails locally on the user's device. This means that once emails are downloaded,

they are usually removed from the server. IMAP4 (Internet Message Access Protocol version 4) Similar to POP3, IMAP is used to access electronic mailboxes (Goralski. 2017; Abdulkadhim and Hayder. 2020). However, IMAP keeps emails on the server, providing more mobility for users to access their emails from different devices. The distinct roles of these protocols in the process of email delivery and retrieval, demonstrating how they contribute to the functioning of email systems (Goralski. 2017).

➤ **TELEtype Network (TELNET)**

Telnet is indeed a two-way text-based communication protocol that establishes a communication channel between two computers. This communication can occur within a Local Area Network (LAN) or remotely via the Transmission Control Protocol (TCP) (Cerf. 2021). Telnet is commonly used by users who need to access applications or data stored on a remote machine, allowing them to interact with the remote system's command-line interface or run applications as if they were directly using the remote machine.

➤ **Simple Network Management Protocol (SNMP)**

Simple Network Management Protocol (SNMP) is accurate. SNMP is indeed a standard Internet protocol designed for monitoring and managing network devices. Commonly supported devices include routers, terminal servers, hosts, and various other network elements. SNMP has evolved over time, with different versions addressing various aspects (Harrington et al. 2002). SNMPv1 laid the foundation by providing the architecture and framework for network management. Subsequent versions, such as SNMPv2c and SNMPv3, have focused on enhancing performance and addressing security concerns. SNMPv3, in particular, introduced advanced security features, making it more robust for managing and monitoring network devices in diverse environments (Hong and Yang. 2013).

❖ **Application layer attacks**

Application-layer attacks, reflecting the diverse range of threats in the cybersecurity landscape. Some notable attacks you mentioned include Denial of Service (DoS), Distributed Denial of Service (DDoS), SMTP attacks, FTP bounce attacks, insecure HTTP, browser hijacking, buffer overflow, malware attacks, and data attacks (Peng et al. 2013; Tripathi and Hubballi. 2021). It's significant that application-layer DoS attacks have gained popularity, and the frequency appears to be on the rise, as you mentioned, doubling every quarter in 2019 (Tripathi and Hubballi. 2021). This underscores the ongoing challenges in securing application-layer services and the need for robust defense mechanisms to protect against these evolving threats. Application-layer security is crucial, considering the variety of attacks that can target vulnerabilities in software and web applications. Various attacks and their potential consequences in the realm of cybersecurity. DoS and DDoS Attacks exhausting the IP address pool and overwhelming servers to disrupt responsiveness. Vulnerabilities and Flaws including DHCP starvation, NTP time-shifting/sticking, and slow HTTP DoS, causing performance degradation. SMTP Attack unauthorized access to client's mail, observing mail addresses, and potential abuse. FTP Bounce Attack similar to SMTP attack, accessing ports using the victim's machine. Browser Hijacking modifying web browser settings through malware. Buffer Overflow Attack overwriting application memory to damage files and expose private information.

Insecure HTTP lack of encryption, making data vulnerable to attacks (Cambiaso et al. 2013; Gonzalez et al. 2014; Mantas et al. 2015; Singh et al. 2018). Malware Attacks including adware, viruses, worms, Trojans, bots, and ransomware.our mention of security mechanisms, such as cryptographic techniques, security features in switches, traffic profiling, machine-learning-based techniques, path and server redundancy, and monitoring traffic features, underscores the importance of a multifaceted approach to cybersecurity. Preventing and mitigating these attacks requires a combination of proactive measures and ongoing vigilance (Peng et al. 2013; Hubballi and Tripathi. 2017; Malhotra et al. 2017; Tripathi and Hubballi. 2017; Dietzel et al. 2018;Tripathi and Hubballi. 2018).

Presentation layer

The Presentation Layer in the OSI model is accurate. The Presentation Layer, as the sixth layer, indeed serves as a bridge between the Application Layer and the Transport Layer. Its primary responsibilities include managing the fields in messages exchanged between end-users and handling the semantics of those fields. Data translating messages between high-level and low-level language data to ensure compatibility between different systems. Encryption–Decryption encrypting plain text messages from the Application Layer for secure transmission and decrypting encrypted data from the Transport Layer at the receiving end. Compression reducing the number of bits in loss-tolerant data to optimize transmission efficiency. In summary, the Presentation Layer plays a crucial role in ensuring that data exchanged between end-users is properly formatted, secure, and efficient in terms of transmission (Kurose and Ross. 2012; Hollis. 1983).

❖ Presentation-layer protocols

Presentation-layer protocols provides a good insight into some of the specific protocols associated with the Presentation Layer.Lightweight Presentation Protocol (LPP) a protocol designed to facilitate the exchange of graphical information between computers. Secure Socket Layer (SSL) Protocol while SSL is commonly associated with the Transport Layer (SSL/TLS), its functions often involve presentation-layer tasks, especially in securing data through encryption and decryption. Apple Filing Protocol (AFP) a proprietary network protocol used by Apple devices for file sharing. Independent Computing Architecture (ICA) a protocol used by Citrix systems for remote application delivery and virtual desktop access. NetWare Core Protocol (NCP) associated with Novell NetWare, NCP facilitates communication between NetWare clients and servers. Network Data Representation (NDR) an RPC (Remote Procedure Call) protocol used in Distributed Computing Environment (DCE) to define how data is represented and transmitted between different architectures (Hollis. 1983; Roberts. 2005). Each of these protocols plays a specific role in supporting data presentation, exchange, and representation in various network and computing environments.

❖ Presentation-layer attacks

The critical role of Secure Socket Layer (SSL) in securing web services, especially in sensitive areas like banking and online shopping. It's crucial for encrypting data during transmission, ensuring the confidentiality and integrity of the information exchanged between clients and servers. Malformed SSL Attacks exploiting vulnerabilities in SSL implementations by sending malformed SSL messages to disrupt communication.

SSL Stripping removing SSL data from client request messages and opening multiple ports for the corresponding server, potentially compromising the security of the communication. Change-Cipher Specification (CCS) Manipulation Attacks manipulating CCS messages improperly during an already opened session, causing passive session issues and negotiation failures (Kumar. 2016; Keerthi. 2016). These attacks underscore the importance of addressing vulnerabilities not only in the SSL protocol itself but also in its implementation to ensure the effectiveness of secure communication over the internet.

Session layer

The Session Layer in the OSI model is accurate. Position in the OSI Model the Session Layer is the fifth layer, positioned between the Presentation Layer and the Transport Layer. Data Management data exchange between end-user applications, facilitating communication in half or full-duplex modes. Communication Sessions the Session Layer is responsible for establishing, managing, and terminating communication sessions between two systems. Synchronization plays a role in synchronizing information from different resources during communication. Checkpoints the Session Layer allows the process of adding checkpoints, ensuring data integrity and reliability during communication. Remote Procedure Calls (RPCs) utilizes RPCs to implement services that control the mentioned functionalities, providing a mechanism for programs to communicate with each other. In summary, the Session Layer is crucial for managing the establishment, maintenance, and termination of communication sessions, contributing to the reliability and synchronization of data exchange between systems (Koksal and Tekinerdogan. 2017).

❖ Session-layer protocols

Remote Procedure Call (RPC) widely used in application environments for session-layer services, RPC facilitates communication between processes on different systems. Session Control Protocol (SCP) a protocol used for controlling sessions, managing communication between systems. Zone Information Protocol (ZIP): Involved in network management, ZIP provides information about the zones in a network, aiding in communication setup. Real-time Transport Control Protocol (RTCP) primarily associated with the transport of real-time multimedia data, RTCP works in conjunction with RTP (Real-time Transport Protocol) to control and provide feedback on the quality of service. Password Authentication Protocol (PAP): Typically used in authentication processes, PAP verifies the identity of user. OSI Session Layer Protocol (ISO-SP, a.k.a. X.225): Part of the OSI networking framework, ISO-SP (or X.225) manages sessions in the OSI model, providing a reliable and structured approach to session-layer communication. Each of these protocols serves specific purposes within the session layer, addressing communication, control, authentication, and network management needs (Salman and Jain. 2017).

❖ Session-layer attacks

Session Hijacking divided into three groups active Session Hijacking attacker directly targets an active session. Passive Session Hijacking attacker positions themselves between the user and server, intercepting and relaying packets to act as a valid user. Hybrid Session Hijacking attacker combines features of active and passive session hijacking based on their goal (Baitha. 2018).

Tools for Session Hijacking include popular tools like Wireshark, T-SightS, Hunt, Hamster, and Ferret. Session ID Stealing Sniffing monitoring victim's network traffic to obtain session IDs from unencrypted packets. Brute Force trying all possible combinations to deduce the session ID (not time-effective). Cross-Site Scripting (XSS): Exploiting vulnerabilities in websites to steal session IDs. Understanding these attack methods and tools is crucial for implementing effective security measures to protect against session-layer vulnerabilities (Baitha. 2018; Jain et al. 2015; Burgers et al. 2013).

Transport layer

The transport layer, located at Layer 4 of the OSI model, is responsible for end-to-end communication and data delivery between devices on a network. Segmentation and Reassembly the transport layer breaks down large messages from the upper layers into smaller segments for efficient transmission. At the receiving end, it reassembles these segments into the original message. Flow Control ensures that the sender doesn't overwhelm the receiver with data. It manages the flow of data to prevent congestion and buffer overflow. Error Detection and Correction implements mechanisms to detect errors in transmitted data and, in some cases, correct them. This enhances the reliability of data delivery. Acknowledgment provides acknowledgment of successfully received data segments. If a segment is not acknowledged, the transport layer initiates the retransmission of that segment. Connection Establishment and Termination in connection-oriented communication, the transport layer establishes, maintains, and terminates connections between devices.

This is crucial for reliable communication. Port Numbers assigns source and destination port numbers to identify specific applications on devices. This allows multiple applications on a device to use the network simultaneously. In summary; the transport layer ensures reliable, error-checked, and orderly delivery of data between applications on different devices across a network. Notable transport layer protocols include TCP (Transmission Control Protocol) and UDP (User Datagram Protocol).

❖ Transport-layer protocols

The Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) are both protocols residing in the transport layer of the OSI model. TCP (Transmission Control Protocol) is a connection-oriented protocol that ensures reliable and ordered delivery of data (Maltz et al. 1998; Kurose and Ross. 2012; Welzl et al. 2021).

It establishes a connection before data transfer, manages acknowledgment and retransmission of lost data, and guarantees the correct sequencing of received data. TCP is commonly used for applications where data accuracy and integrity are crucial, such as file transfers, web browsing, and email. UDP (User Datagram Protocol), on the other hand, is a connectionless protocol. It does not establish a connection before sending data and does not guarantee reliable delivery. UDP is often chosen for applications where speed and efficiency are more critical than error checking and retransmission.

Examples include real-time applications like online gaming, streaming, and VoIP (Voice over Internet Protocol). In summary, TCP prioritizes data quality and reliability, making it suitable for applications that require accurate and complete data delivery. UDP prioritizes speed and efficiency, making it suitable for applications where occasional data loss is acceptable. The choice between TCP and UDP depends on the specific requirements of the application.

❖ **Transport-layer attacks**

TCP Flooding Attack (TCP SYN Attack) is a type of DDoS (Distributed Denial of Service) attack where an attacker floods a target system with a large number of TCP SYN (synchronization) packets. The goal is to overwhelm the system and exhaust its resources, making it unable to handle legitimate connections. UDP Flooding Attack in this DDoS attack, the attacker spoofs the victim's IP address and sends a high volume of UDP (User Datagram Protocol) packets to the target system. The intention is to disrupt the victim's computer system and weaken its performance (Schuba et al. 1997; Eddy. 2007). TCP Sequence Prediction Attack involves predicting the sequence numbers used to identify packets in a TCP connection. By successfully predicting these numbers, an attacker can send counterfeit packages to the victim's computer, potentially compromising the integrity and security of the communication. Mitigating these threats often involves implementing security measures such as firewalls, intrusion detection/prevention systems, and encryption protocols to protect against unauthorized access, traffic anomalies, and data interception (Pandey and Saini. 2014; Sula. 2018). Additionally, proper network configuration and monitoring can help identify and respond to suspicious activities (Bellovin.1989; Qian et al. 2012).

Network layer

The network layer, positioned at Layer 3 of the OSI model, plays a crucial role in routing data packets between devices across different networks (Kurose and Ross. 2012). Logical Addressing assigns logical addresses (such as IP addresses) to devices for network identification. These addresses are used for routing data across networks. Routing determines the best path for data packets to reach their destination. Routers at the network layer make forwarding decisions based on logical addresses. Packet Forwarding divides messages into packets and manages their transportation across the network. Each packet contains source and destination logical addresses. Fragmentation and Reassembly if necessary, the network layer can fragment large packets into smaller fragments for transmission. At the receiving end, these fragments are reassembled into the original packets. Internet Protocol (IP) primary protocol at the network layer is IP. It provides the addressing and routing functions necessary for data communication across interconnected networks. Logical-Physical Address Mapping network layer maps logical addresses to physical addresses, enabling the delivery of data packets to specific devices on a network. TTL (Time-to-Live) a field in the IP header that limits the lifespan of a packet, preventing it from circulating endlessly in the network. Sub-netting allows the division of a larger network into smaller, more manageable subnetworks, enhancing network efficiency and security (Akin and Korkmaz. 2019). In summary, the network layer facilitates the movement of data across different networks, ensuring proper addressing, routing, and delivery of packets. It acts as a bridge between the data link layer below it and the transport layer above it in the OSI model.

A packet is forwarded from Host A to Router 1 (R1) and then to Host B via Router 2 (R2), is indeed part of the data plane in networking. The data plane, also known as the forwarding plane, is responsible for the actual forwarding of data packets. It involves the mechanisms and decisions made by networking devices, such as routers and switches, to direct packets from their source to their destination based on the information in the packet headers. Forwarding from Host A to R1 the data plane in R1 is responsible for processing the packet and determining the appropriate next-hop, which is Router 2 (R2) in this case.

Forwarding from R1 to R2 the data plane in R2 then takes over, making decisions on how to forward the packet toward its final destination, Host B (**Figure 7**). Understanding and managing the data plane is crucial for ensuring efficient and reliable packet forwarding in a network (Akin and Korkmaz. 2019). This is in contrast to the control plane, which involves the higher-level decisions about how data should be forwarded and the overall management and configuration of the network devices.

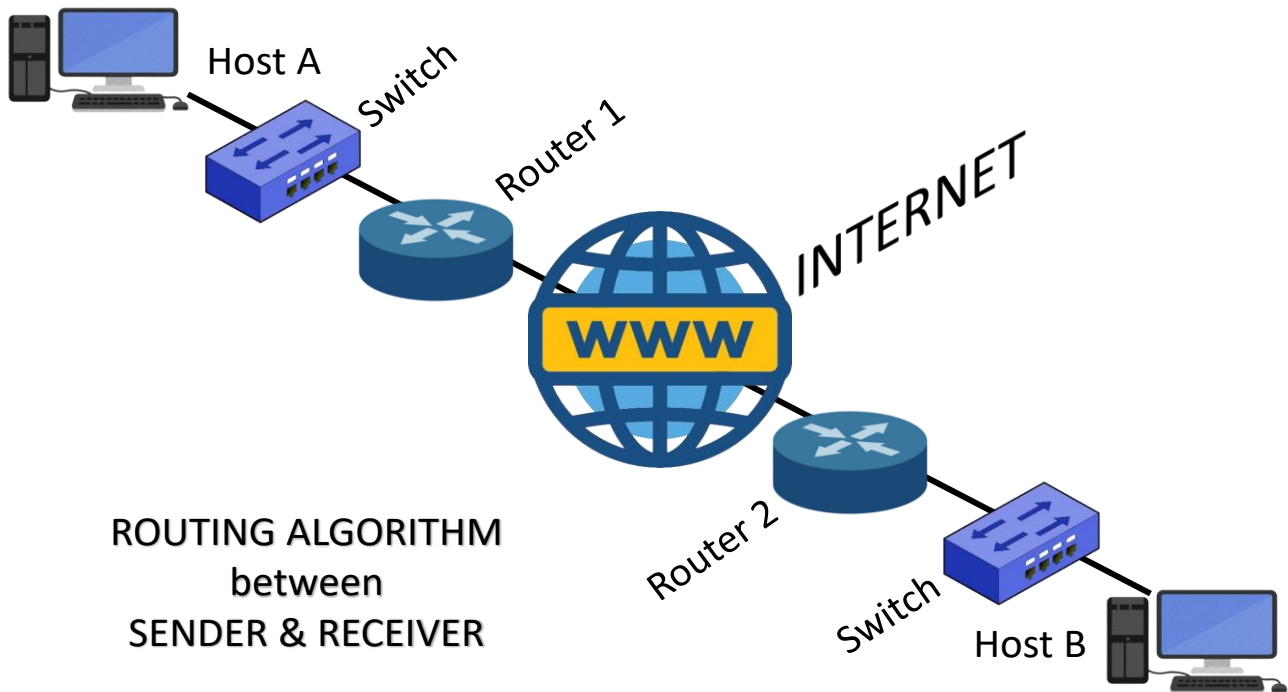


Figure 7. Routing algorithm between sender and receiver

❖ Network-layer protocols

IPv4 (Internet Protocol version 4) is the fourth version of the Internet Protocol and is the most widely used for internet traffic. IPv4 uses 32-bit addresses, allowing for approximately 4.3 billion unique addresses. However, due to the rapid growth of the internet, IPv4 address exhaustion has become a concern, leading to the development and adoption of IPv6. IPv6 (Internet Protocol version 6) is the successor to IPv4 and was designed to address the limitations of IPv4, particularly the exhaustion of available addresses. IPv6 uses a 128-bit address scheme, providing an enormous number of unique addresses, ensuring the continued growth of the internet.

ICMP (Internet Control Message Protocol) are a network layer protocol used for sending error messages and operational information about network conditions. It is often employed by network devices to communicate error messages, such as unreachable hosts or network congestion, and is integral to the functioning of IP networks. These protocols, along with others, form the backbone of internet communication, enabling the routing and delivery of data packets across diverse networks.

❖ **Network-layer attacks**

Smurf Attack is a type of DDoS attack where an attacker spoofs the IP address of a victim's computer and sends forged ICMP echo requests to a broadcast address. The devices in the same broadcast domain reply with ICMP echo replies, causing a flood of traffic directed at the victim, potentially leading to network congestion and denial of service (Sajjad. 2009; Bouyeddou et al. 2018). In an IP spoofing attack, the attacker uses a trusted IP address to send packets to a server. The server, unable to identify the true source of the packets, may accept them as legitimate.

This can be exploited by attackers to gain unauthorized access to the server. In a hijacking attack, an attacker disrupts a server-client session and establishes a new session between themselves and the server (Zou et al. 2016). This can lead to unauthorized access, data interception, or manipulation of the communication between the legitimate parties. The importance of implementing security measures such as firewalls, intrusion detection systems, and encryption protocols to protect against unauthorized access, data manipulation, and service disruptions. Regular monitoring and updates to security protocols are essential to mitigate the risk of network-layer attacks (Manivannan and Sathiyamoorthy. 2014).

Data link layer

The data link layer is indeed the second layer of the OSI model, situated between the physical layer and the network layer. Protocols and Nodes the data link layer involves protocols that operate on nodes, which can be various networking devices such as hosts, routers, switches, and Wi-Fi access points. Communication Channels (Links) nodes communicate over communication channels commonly referred to as links. These links can take various forms, including wired connections (e.g., Ethernet) or wireless connections (e.g., Wi-Fi).

Transfer of Data: The primary function of the data link layer is to facilitate the reliable transfer of data between the source host and the destination host over the communication links. This layer is responsible for framing, addressing, and error detection within the data transmitted on a link. Understanding the data link layer is essential for managing the flow of data within a local network, ensuring proper addressing and error handling as data moves between devices (Kurose and Ross. 2012).

❖ **Data link layer protocols**

Address Resolution Protocol (ARP) operates at the data link layer, even though it's often associated with the network layer. ARP is responsible for mapping network layer addresses (like IP addresses) to link-layer addresses (like MAC addresses) (David. 1982). It helps devices on a local network find each other's hardware addresses. Spanning Tree Protocol (STP) is crucial for preventing loops in network topologies. By exchanging Bridge Protocol Data Units (BPDU) among devices, STP identifies and eliminates redundant paths, ensuring a loop-free and stable network. Serial Line Internet Protocol (SLIP) is a protocol that encapsulates Internet Protocol (IP) packets over serial ports and modem connections (Mahmood et al. 2020). While SLIP was widely used in the past for point-to-point communication over serial links, it has largely been superseded by more advanced protocols like Point-to-Point Protocol (PPP). These protocols play key roles in addressing, topology stability, and serial communication at the data link layer, contributing to the effective functioning of local and wide-area networks (Stevens. 1994).

❖ **Data-link-layer attacks**

MAC Attacks (CAM Table Flooding) attacks force a switch to operate like a hub by flooding the Content Addressable Memory (CAM) table, causing the switch to forward packets to all ports. This degrades network performance and can be exploited for unauthorized access. In STP attacks; forged Bridge Protocol Data Unit (BPDU) messages are used to manipulate the Spanning Tree topology.

By inducing frequent topology changes, attackers may disrupt the network or launch a Denial of Service (DoS) attack (Mahmood et al. 2020). ARP Poisoning Attacks (MAC Spoofing) poisoning, also known as MAC spoofing, occurs at the data link layer even though ARP operates at the network layer. Attackers send gratuitous ARP (GARP) packets, announcing spoofed MAC and IP addresses. Devices on the local network update their caches accordingly, allowing the attacker to intercept or manipulate traffic by forcing it through their switch. The importance of implementing security measures such as port security, BPDU guard, and ARP spoofing detection to safeguard the data link layer. Protecting against these threats helps maintain the integrity and security of local networks (Tripathi and Mehtre. 2014; Mahmood et al. 2020).

Physical layer

The physical layer is the bottommost layer in the OSI model, serving as the foundation for the higher layers. Bit Transmission unlike the data link layer, which deals with frames, the physical layer is concerned with the transmission of individual bits over the physical medium, such as cables, fibers, or wireless channels. Responsibilities the primary responsibilities of the physical layer include the conversion of bits into electrical or optical signals for transmission and the reception of signals, converting them back into bits. It also manages characteristics such as voltage levels, physical connectors, and modulation schemes. Medium Independence the physical layer is often considered medium-independent, meaning it defines the characteristics of the hardware but doesn't prescribe the specific medium or technology. It adapts to various physical transmission mediums. In summary, the physical layer is responsible for the actual transmission of raw data bits over the physical medium, providing the electrical, mechanical, and procedural specifications for this transmission. It forms the base upon which the higher layers build their more complex communication processes.

❖ **Physical layer protocols**

The protocols at the physical layer are highly dependent on the characteristics of the transmission medium or material used for communication links. This layer deals with the physical characteristics of the transmission medium and the means by which bits are transmitted. Protocols and technologies at the physical layer for fiber optics may involve the modulation of light signals for data transmission. Twisted-Pair Copper Wire for copper-based connections, protocols at the physical layer may involve electrical signaling using techniques like amplitude modulation or phase modulation. Wi-Fi is wireless communication, such as Wi-Fi, relies on protocols at the physical layer that define how radio frequency signals are modulated and transmitted to carry data wirelessly. The physical layer specifications ensure that devices at either end of a communication link can understand how to encode, transmit, and decode bits based on the specific characteristics of the transmission medium in use. This adaptability to various media is a key feature of the physical layer in the OSI model.

❖ **Physical-layer attacks**

In a jamming attack, a malicious node intentionally interferes with the transmission of packets, disrupting the normal communication and reducing network performance. This type of attack is particularly impactful in Wireless Sensor Networks (WSNs) where communication channels are crucial for sensor data (Liu. 2012). Wiretapping involves an attacker connecting a third-party wire to the communication lines. This allows the attacker to observe and analyze the flow of data, either in active mode (capturing data actively transmitted) or passive mode (simply monitoring without altering the data). Wiretapping is a significant security concern, especially in wired communication. Tampering is an unauthorized act where an attacker modifies a system, its components, or data on the network (Deka. 2014). This could involve altering hardware components or manipulating data in transit. Tampering attacks pose a serious threat to the integrity and reliability of a network. Protecting against these physical-layer attacks involves implementing security measures such as encryption for data confidentiality, shielding against electromagnetic interference, and physical security measures to prevent unauthorized access to communication lines.

Threats

The increasing reliance on wireless communications introduces a range of security challenges due to the proliferation of devices (David. 1982; Stevens. 1994; Aliu et al. 2012; Liu. 2012; Deka. 2014; Manivannan and Sathiyamoorthy. 2014; Tripathi and Mehtre. 2014; Zou et al. 2016; Mahmood et al.; 2020). Common threats include cyber criminal activities, malicious attacks, data forging, identity theft, and more, particularly with the widespread use of smartphones for online banking and personal accounts (Zou et al. 2016). To address these concerns, enhancing wireless communication security is crucial.

Wireless security attacks

In malicious association attack, the malicious actor targets a company's access point and manipulates their cracked laptop's wireless network card to mimic a legitimate access point. When a user unknowingly connects to this deceptive access point, the attacker can potentially steal the user's password. Moreover, this attack can be a gateway for the attacker to launch further attacks on the wired network. Accidental Association threat involves an unauthorized user unintentionally connecting to a wireless access point belonging to a nearby company's network. As a result, sensitive company information may be exposed, posing a security issue due to the inadvertent connection of an unauthorized user. Implementing strong encryption protocols (e.g., WPA2, WPA3) helps protect against password theft and unauthorized access. Utilizing secure authentication methods, such as WPA2-Enterprise with 802.1X, enhances access control. Regularly monitoring the wireless network for unusual activities or unauthorized devices helps detect and respond to potential threats (Choi et al. 2008; Aliu et al. 2012; Stallings et al. 2012; Elsayy et al. 2013; Kavianpour and Anderson. 2017). Educating users about the importance of verifying network connections and being cautious about connecting to unknown networks helps prevent accidental associations. Employing IDS can aid in identifying and responding to malicious associations and other suspicious activities (**Figure 8**).

Security concerns within ad hoc networks and nontraditional networks like Bluetooth. Ad hoc networks are Peer-to-Peer (P2P) networks where clients connect wirelessly without relying

on a centralized access point (Choi et al. 2008; Aliu et al. 2012; Stallings et al. 2012; Elsaywy et al. 2013; Kavianpour and Anderson. 2017). Ad hoc networks often lack robust protection, making them susceptible to a range of security threats. Nontraditional networks like Bluetooth face security risks due to weaker security measures. Attackers may exploit the weak security by performing MAC spoofing, identifying MAC addresses and compromising the network (Pham et al. 2010; Wadhwa and Arora. 2017; Al-Rimy et al. 2018; Chen et al. 2019). The Caffé Latte attack specifically targets the WEP (Wired Equivalent Privacy) security protocol. Attackers exploit weaknesses in WEP by obtaining the key through a flood of encrypted ARP responses generated using ARP requests. An attack where an unauthorized entity intercepts and potentially alters communication between two parties (Choi et al. 2008; Aliu et al. 2012; Stallings et al. 2012; Elsaywy et al. 2013; Kavianpour and Anderson. 2017). An attack that disrupts or makes a network resource unavailable to users. Unauthorized injection of data into a network, often to exploit vulnerabilities or manipulate traffic (Zou et al. 2016).



Figure 8. Wireless network

Securing wireless networks

Built-in encryption mechanisms in wireless routers and access points provide a fundamental layer of protection for data transmitted over the network. Encryption is a powerful tool to safeguard data confidentiality and prevent unauthorized access. Utilizing security software and keeping it updated helps protect against various types of malware and unauthorized access attempts. Frequent updates ensure that the security software is equipped to deal with the latest threat (Zou et al. 2016; Choi et al. 2008; Aliu et al. 2012; Stallings et al. 2012; Elsaywy et al. 2013; Kavianpour and Anderson. 2017). Changing default router identifiers and passwords is crucial since attackers often know these defaults. Using unique and strong credentials enhances security by preventing unauthorized access.

Disabling the broadcast of network identifiers contributes to security by making it more challenging for attackers to discover the network. Restricting network access based on MAC addresses adds an additional layer of security (Goudarzi et al. 2022). While helpful, this method has limitations as attackers can attempt to mimic MAC addresses. Educating users about the significance of network security helps create a security-conscious user base. Routinely using security measures becomes a habit and reduces the likelihood of successful attacks. By combining these measures, organizations and individuals can significantly enhance the security of their wireless networks, creating a more resilient defense against potential threats and attacks (Goudarzi et al. 2022).

5. EVALUATION OF CYBER SECURITY VULNERABILITIES, THREATS, ATTACKS, SOLUTIONS, AND FUTURE CHALLENGES

It's crucial to recognize the dual role of technical and nontechnical solutions in addressing cyber security threats. Nontechnical measures, such as physical security and administrative policies, play a vital role alongside technical solutions like cryptography, access control, and leveraging technologies like big data for threat analysis (Figure 9).

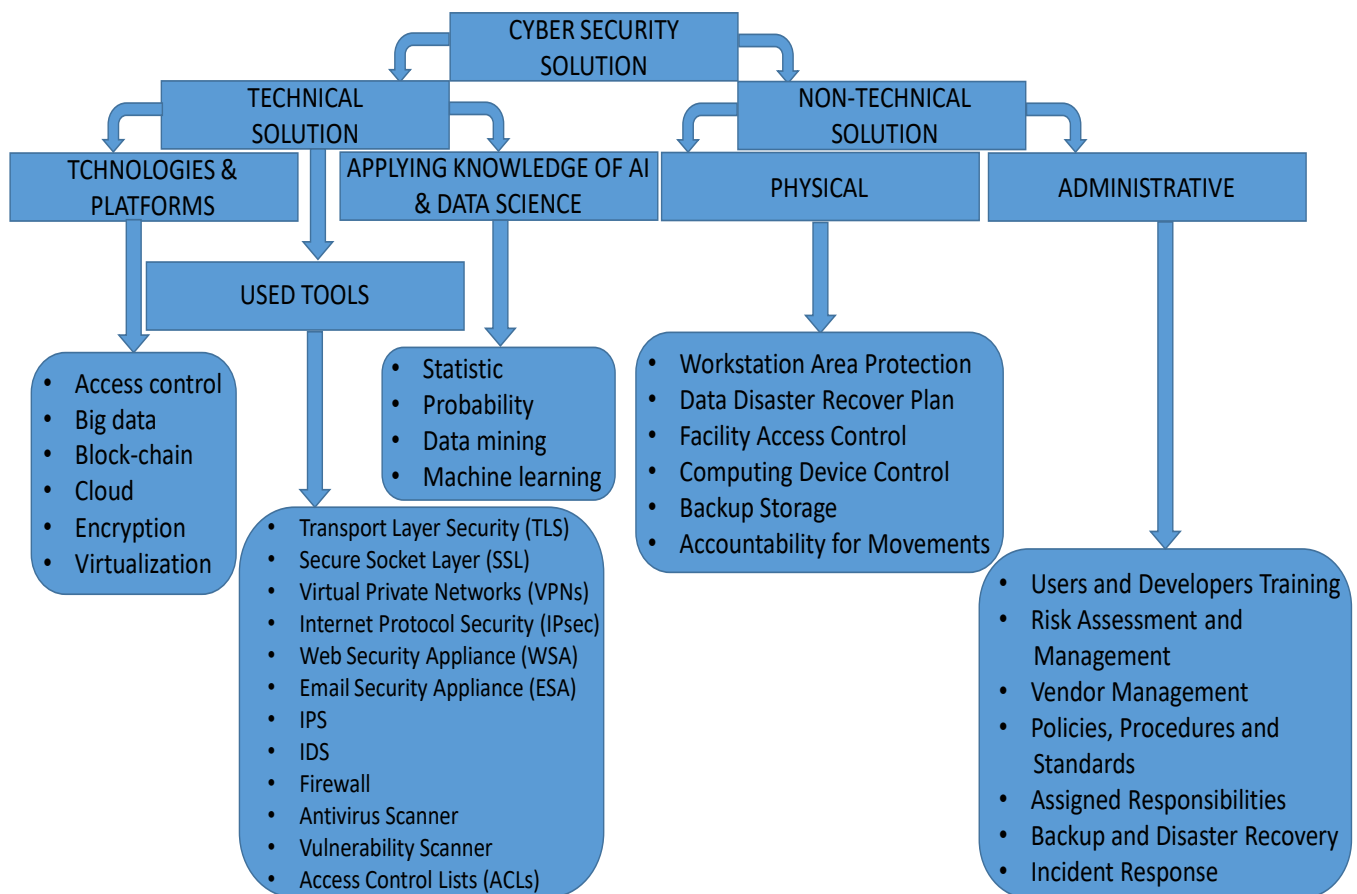


Figure 9. Cyber security solution

Emerging technologies like blockchain, virtualization, and big data contribute significantly to enhancing cyber security. Blockchain's ability to validate data consistency and detect complex attacks adds an additional layer of security. Virtualization, by separating software applications from hardware, not only improves usability and reduces costs but also minimizes downtime during cyber attacks. Meanwhile, cloud computing offers proactive threat management, advanced data security, scalability, high availability, and efficient data recovery, making it a robust platform for addressing cyber security challenges. The integration of these technologies reflects the dynamic landscape of cyber security measures.

The array of tools and protocols you mentioned, including ACLs, firewalls, IDSs, IPSs, VPNs, IPsec, TLS, SSL, WSA, ESA, antivirus scanners, and vulnerability scanners, collectively form a comprehensive defense against cyber threats. Implementing these measures across different layers is crucial for effective protection. Moreover, the incorporation of advanced systems using statistics, probability, data mining, and machine learning techniques represents a proactive approach to detect and counter evolving cyber threats, highlighting the ongoing need for innovation in the field of cyber security.

The dynamic nature of cyber threats requires constant adaptation in detection systems. Incorporating data mining and machine learning techniques introduces a proactive dimension, allowing systems to uncover patterns and adapt to evolving attack strategies. While statistics and probability have long been employed in cyber security, recent years have seen a surge in the popularity of data mining and machine learning. These innovative technologies not only enhance existing detection systems but also imbue them with increased intelligence, enabling more effective responses to contemporary cyber attacks. The evolving landscape underscores the importance of staying at the forefront of technological advancements in cybersecurity.

Machine learning encompasses diverse techniques such as regression, probabilistic methods, distance-based learning, decision trees, dimensionality reduction algorithms, and boosting—bagging algorithms—all valuable in cyber security. These ML techniques contribute to scanning for data breaches and vulnerabilities across computer systems and communication networks (Houichi et al. 2021). Their ability to analyze large datasets quickly and adapt without domain expert input is a key advantage. Moreover, ML methods enhance threat detection accuracy through heuristic techniques, playing a versatile role in various areas within the cyberspace landscape.

The application areas of machine learning in cybersecurity are extensive and encompass:

- - Malware detection (Shah et al. 2022);
- - Spam identification;
- - Fraud detection (Malik et al. 2022)
- - Malicious JavaScript detection;
- - Anomaly detection (Hu et al. 2022; Gadal et al. 2022);
- - Malware classification (Akhtar and Feng. 2022);
- - Risk classification;
- - DNS classification;
- - Phishing detection (Mughaid et al. 2022);
- - Leakage detection;
- - Hidden channel detection;
- - Botnet detection;

- - DDoS detection (Alashhab et al. 2022; Saghezchi et al. 2022; Almaraz-Rivera et al. 2022);
- - Zero-day detection;
- - APTs detection;
- - Social media attacks detection;
- - Vulnerability detection;
- - Cryptographic attack detection.

These diverse applications showcase the versatility of machine learning in addressing various cyber security challenges. The field of cyber security faces a multitude of challenges, ranging from the time-intensive nature of designing secure systems to the evolving landscape of intelligent and automated attacks. The interplay of factors such as human aspects, machine learning limitations, and the complex nature of attack vectors further complicate the efforts to enhance cybersecurity measures.

The intricate nature of cyber security, from time-intensive system design to the evolving landscape of attacks, incorporating human factors, automation, machine learning complexities, and a range of emerging threats (183). Addressing these challenges requires a holistic approach that combines technological advancements, human awareness, and adaptive strategies to stay ahead of evolving threats.

6. CONCLUSIONS

The structured division of the comprehensive review paper into three sections—cyber security fundamentals, threats and vulnerabilities, and network security level by level—provides a systematic and in-depth exploration of the diverse facets of cybersecurity. This approach allows for a thorough examination of the key components and enables the formulation of comprehensive solutions to address the intricate challenges within the realm of cybersecurity. The cyber security fundamentals section provides a nuanced perspective, encompassing both technical and non-technical factors contributing to the surge in cyber attacks. The dynamic nature of cyber threats affects computer-based systems and the internet environment, with changing attack formats and target audiences.

The migration of social activities to the online realm amplifies the impact of cyber attacks. Factors such as software vulnerabilities, network protocol shortcomings, device proliferation, and the complexity of critical systems heighten security risks. Additionally, the virtualization of social life, extensive use of social networks, growing attacker expertise, and user negligence in internet usage further escalate security concerns in the digital landscape.

The section on threats, exploits, and attacks delves into the components of attack strategies, emphasizing the exploitation of system and third-party software vulnerabilities. Cyber attackers leverage well-prepared malicious code blocks to target vulnerabilities in computer-based systems. Common threats and attacks encompass various forms of malware (viruses, worms, rootkits, ransomware, etc.), hacking tools, application attacks, access attacks, cryptography attacks, and Advanced Persistent Threats (APTs). Notably, the emergence of cyber-attacks-as-a-service tools has expanded the landscape, enabling a broader range of attacks and posing challenges to the detection processes.

The comprehensive exploration of network security includeincludee a detailed analysis of common attacks at each layer, along with potential solutions.

Attacks often target specific layer protocols, categorizing them into physical-layer attacks (sniffing), data-link-layer attacks (spoofing: MAC, ARP, etc.), network-layer attacks (man-in-the-middle), transport-layer attacks (reconnaissance), session-layer attacks (hijacking), presentation-layer attacks (phishing), and application-layer attacks (exploits). The success of these attacks is rooted in vulnerabilities within network protocols and misconfigurations of network devices. Protocols such as ARP, DNS, DHCP, FTP, ICMP, IP, TCP, and UDP are susceptible to exploitation. For instance, IP protocol lacks a structure to ensure the accuracy and confidentiality of transmitted packets, leading to potential exposure and alterations. Addressing these issues involves reducing protocol vulnerabilities, introducing new protocols, and ensuring correct and comprehensive configuration of network devices to safeguard data traversing computer networks.

Effectively protecting computer-based systems from attackers necessitates collaboration among individuals, organizations, software developers, and countries, emphasizing both technical and non-technical solutions. In the realm of cyber attacks, non-technical solutions play a crucial role, encompassing administrative-based management, policies, standards, procedures, risk assessment, vendor management, assigned responsibilities, and training. Non-technical concepts are pivotal; even with a meticulously designed protection system by cybersecurity specialists, adequate security remains elusive without comprehensive user training. The synergy of technical and non-technical measures is essential for creating a robust defense against evolving cyber threats.

Technical solutions leverage technological advancements and scientific principles to develop intelligent applications for countering attackers. These solutions can be categorized into three groups: technologies and platforms, utilized tools, and the application of artificial intelligence (AI) and data science. Safeguards data integrity and confidentiality. Restricts data access, enhancing security levels. Enables analysis of vast datasets to uncover unknown patterns and malicious attack features.

Separate software applications from hardware, enhancing usability and reducing costs and downtime during cyber attacks. Provides proactive threat management, advanced data security, scalability, high availability, and efficient data recovery. Assists in validating data consistency and detecting complex attacks. Aid in interpreting and uncovering patterns in data.

Discovers and extracts unknown patterns in large datasets. Enhance attack detection systems by adding new features, improving intelligence against recent cyber threats. The integration of these technical solutions plays a pivotal role in fortifying the resilience of computer-based systems against a diverse range of cyber threats.

Despite the significant strides in technical advances, challenges persist in effectively detecting new and complex cyber attacks.

The escalating complexity of attacks, automation through cyber-attacks-as-a-service, intelligent attacks evading detection systems, biases in machine learning (ML) algorithms assuming data, the difficulty of classifying millions of network connections, handling high-dimensional data, protecting multiple components, and the persistent human factor in security issues all contribute to the ongoing challenges in cybersecurity. Addressing these complexities requires continuous innovation, interdisciplinary approaches, and a concerted effort to stay ahead of the evolving threat landscape.

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